

Learning and Behavior

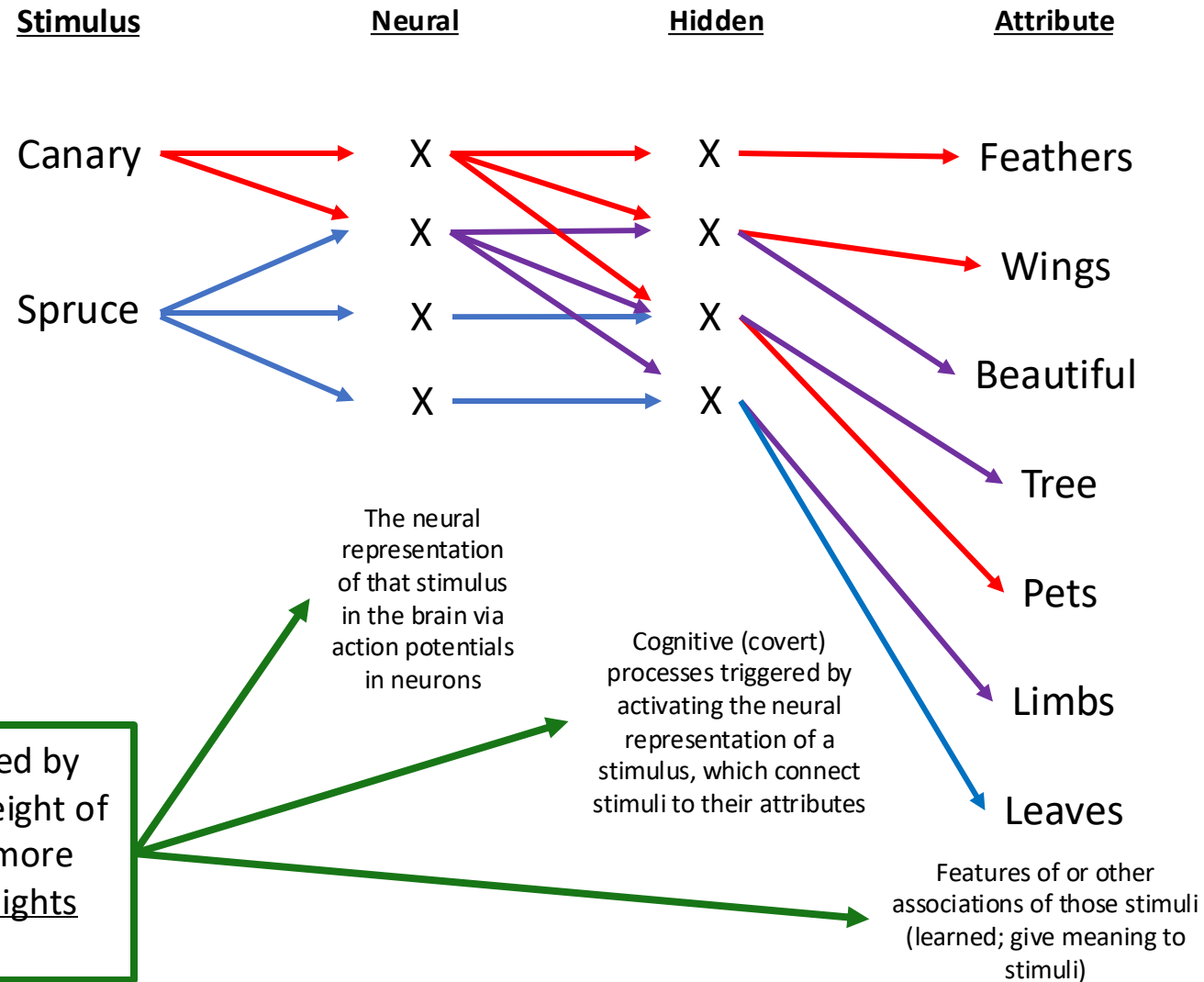
Learning & Behavior (PSY/NSC 3320)

Jean P. Bodet III, Ph.D.

Baylor University

THE ROAD SO FAR

The Connectionist Model



All of these relationships are mediated by **weighted connections** – stronger weight of connection between two concepts, more likely that one triggers the other; weights can change based on experience

NOW

The Neuroscience of Learning: Hebbian Plasticity

Biologically, long-term memory might be described as...

Preserving a record of which neurons represented some experience
(i.e., which neurons were involved in processing/making sense of that experience)

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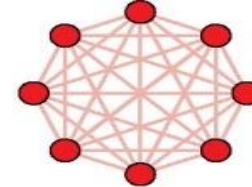
Hebbian Plasticity:

The biology of *all* (implicit
and explicit) learning...

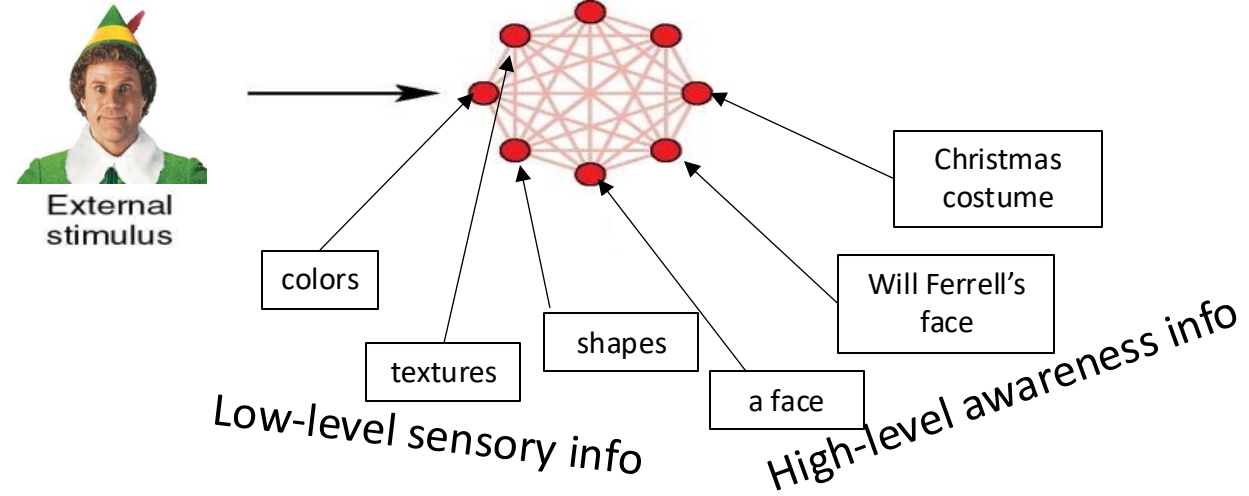
First presentation of stimulus → Neural representation



External
stimulus



First presentation of stimulus → Neural representation

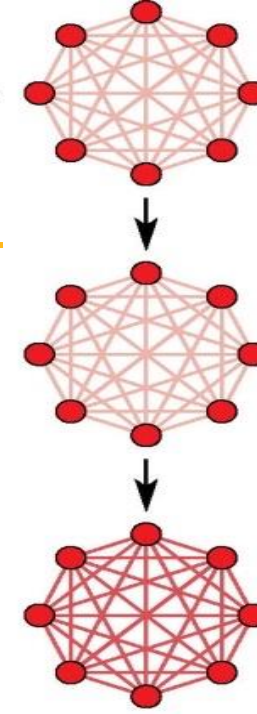


Each node is a neuron / neurons that are active anywhere in the brain, representing some aspect of that stimulus

First presentation of stimulus → Neural representation



Fire together → Wire together

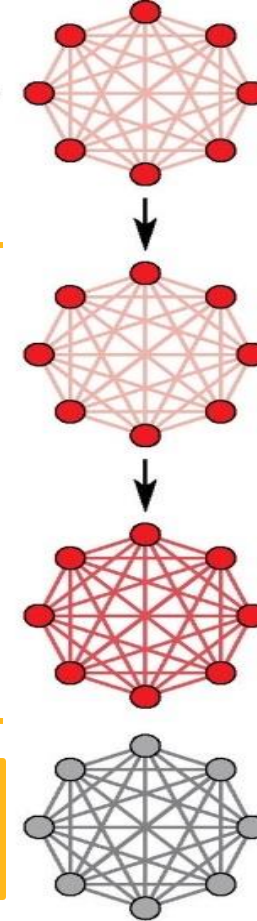


First presentation of stimulus → Neural representation



Fire together → Wire together

Wires (strengthened connections)
remain after firing stops

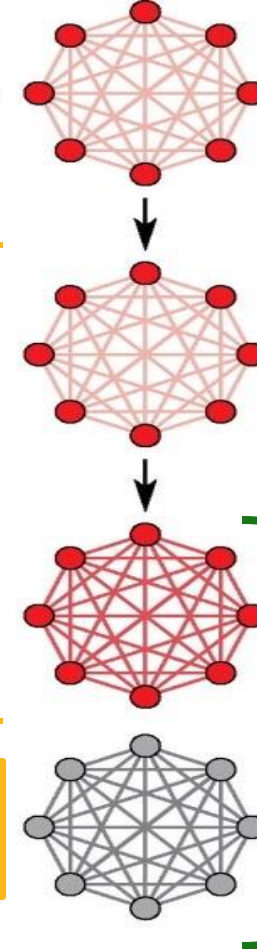


First presentation of stimulus → Neural representation



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CELL ASSEMBLY
/
MEMORY ENGRAM
The learned (stored)
neural representation of
an external stimulus

First presentation of stimulus → Neural representation



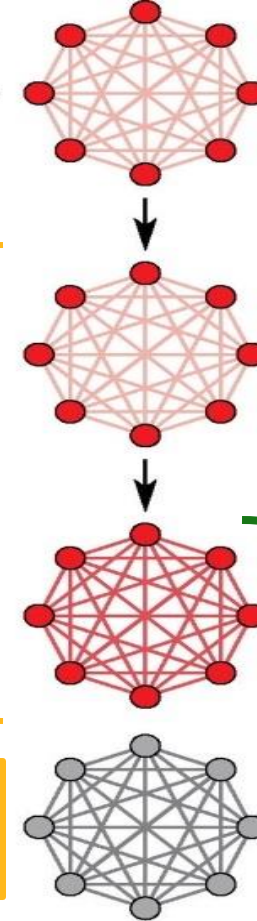
The thickening of these lines
representing the
strengthened relationship
between two pieces of info
reminds me of something
we've seen before...



External
stimulus

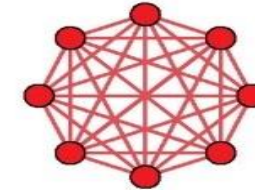
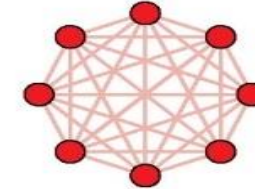
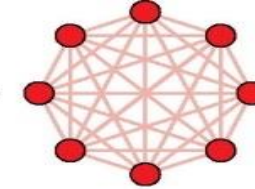
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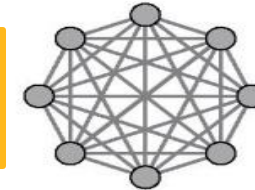
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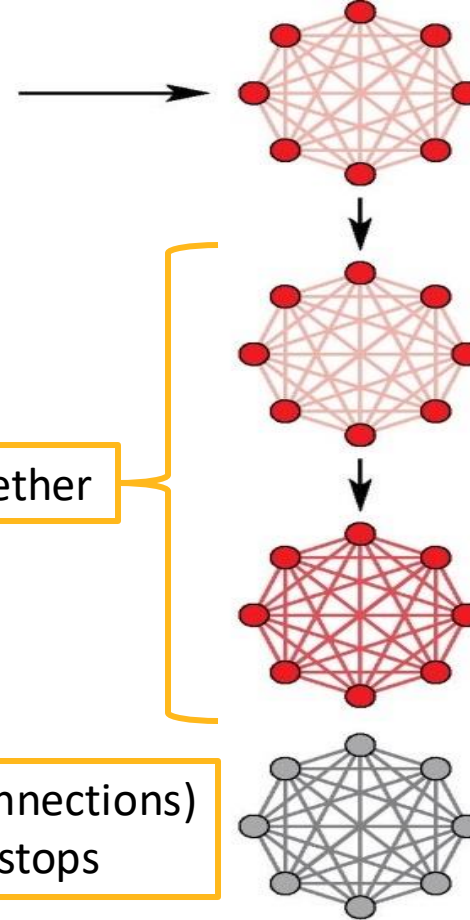
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TIME PASSES —————

First presentation of stimulus → Neural representation

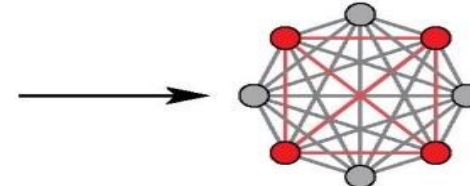


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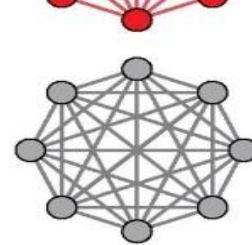
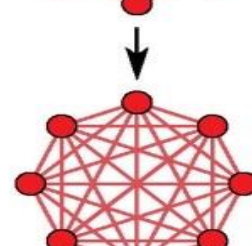
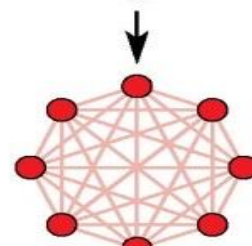
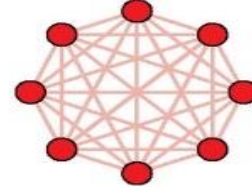
Wires (strengthened connections)
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TIME PASSES

Later, whole or partial stimulus



First presentation of stimulus → Neural representation

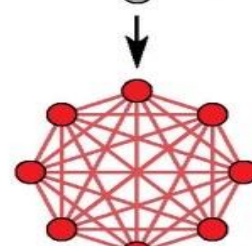
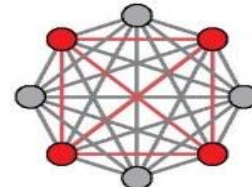


Fire together → Wire together

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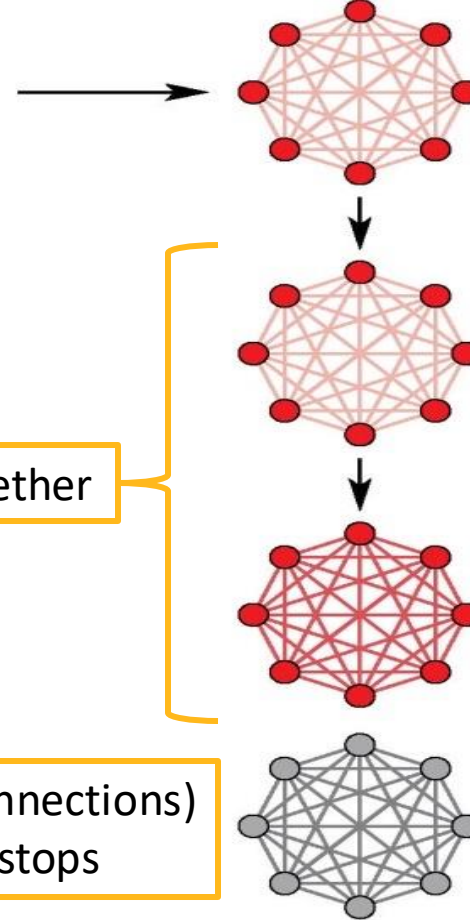
TIME PASSES

Later, whole or partial stimulus



Reactivates whole cell assembly

First presentation of stimulus → Neural representation



Two halves to this phenomenon:

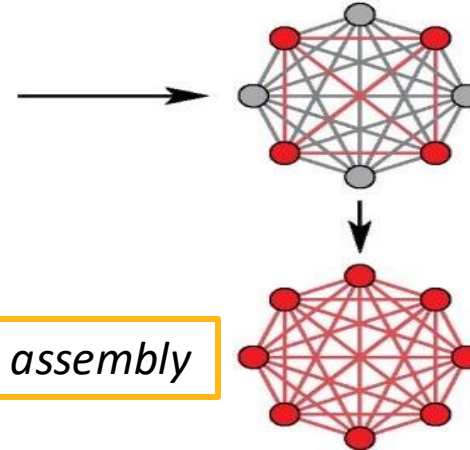
Systems Consolidation
Synaptic Consolidation

Fire together → Wire together

Wires (strengthened connections)
remain after firing stops

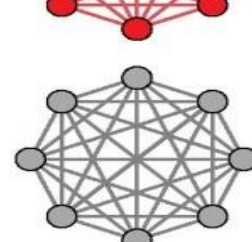
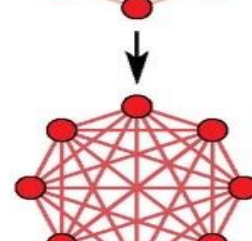
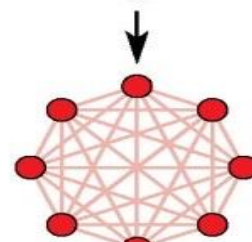
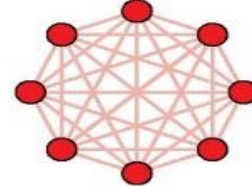
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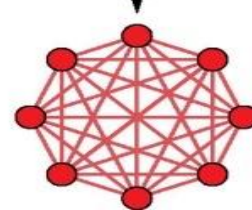
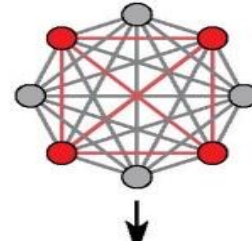
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What is “consolidation?”

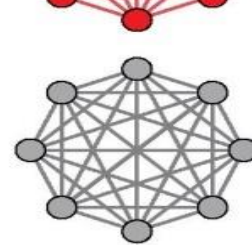
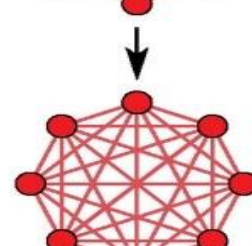
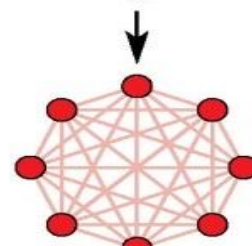
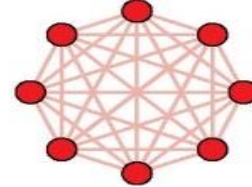
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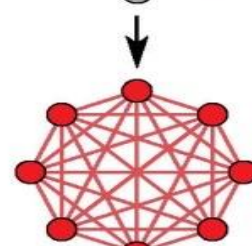
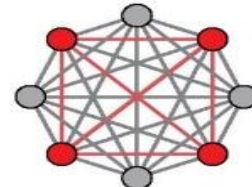


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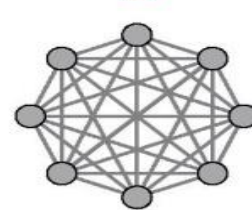
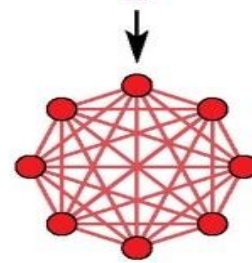
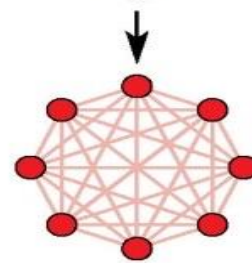
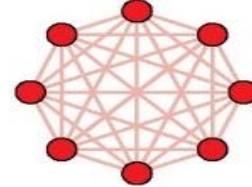
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Later, whole or partial stimulus



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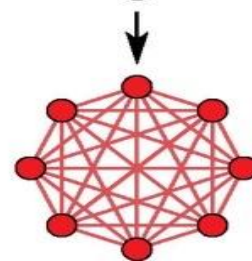
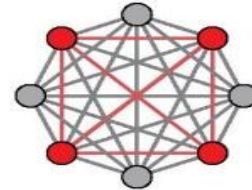


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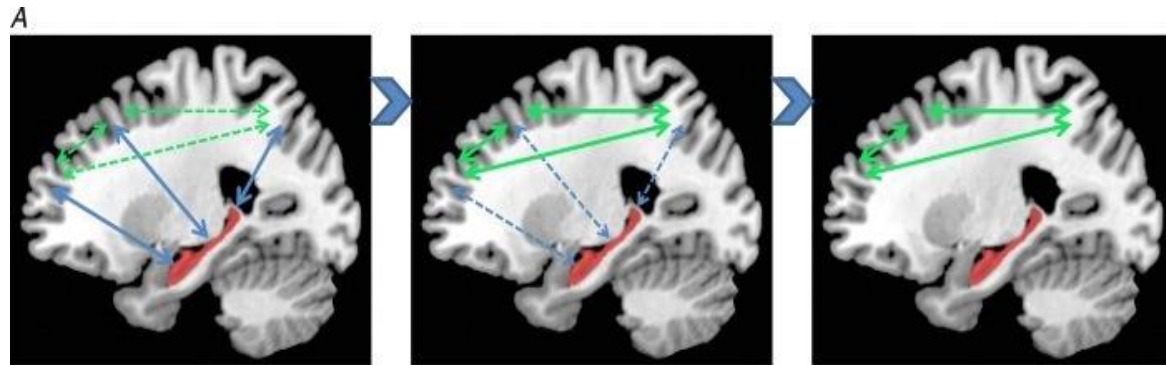
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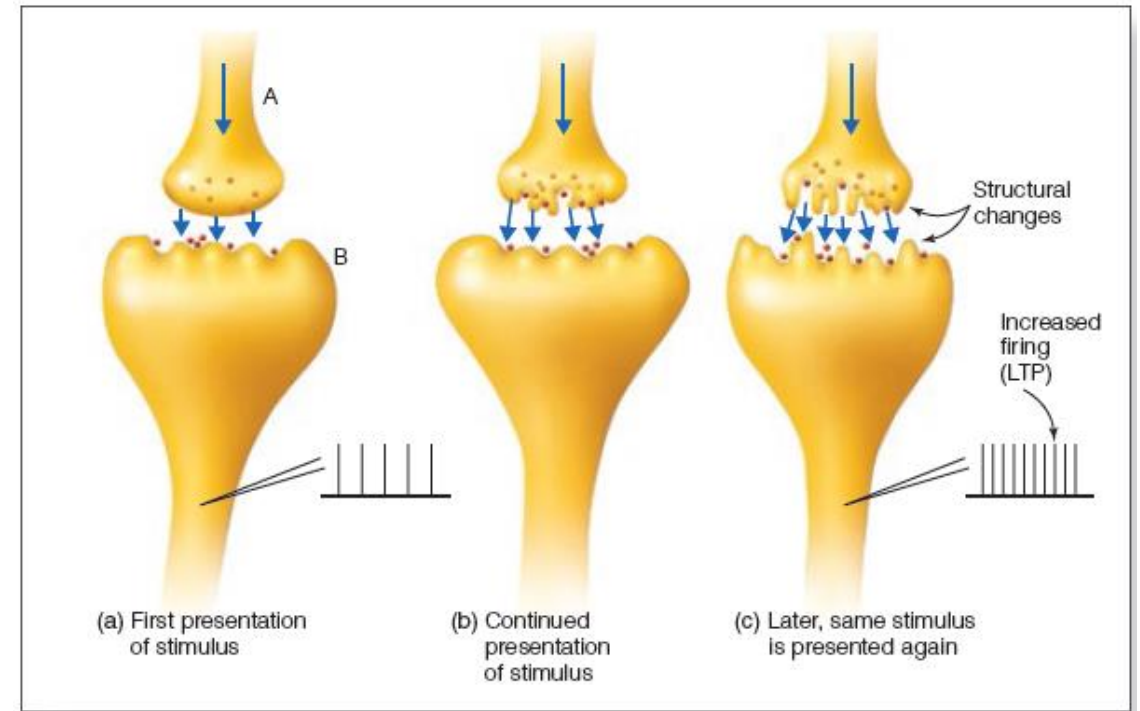


Reactivates whole cell assembly

Systems Consolidation



Synaptic Consolidation

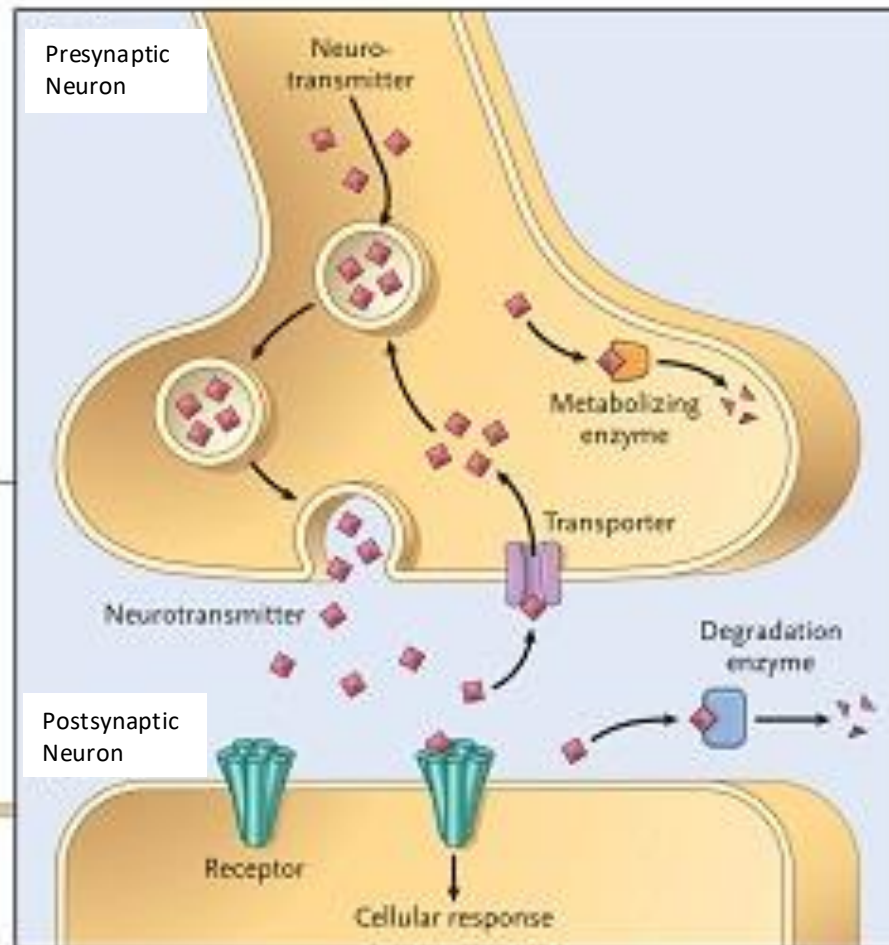
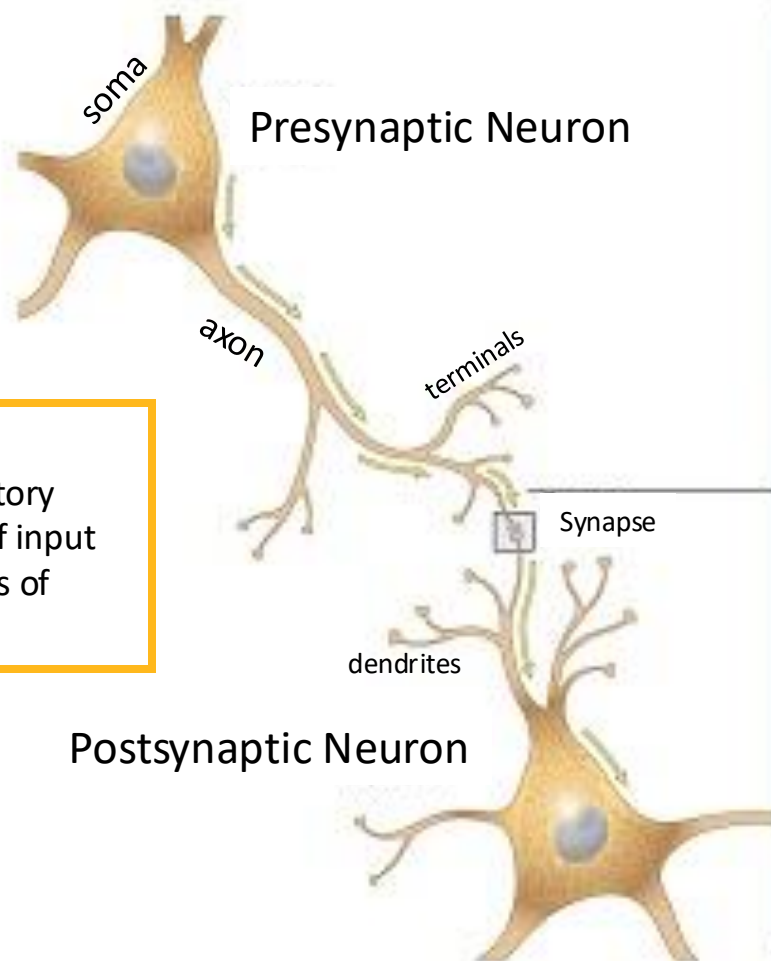


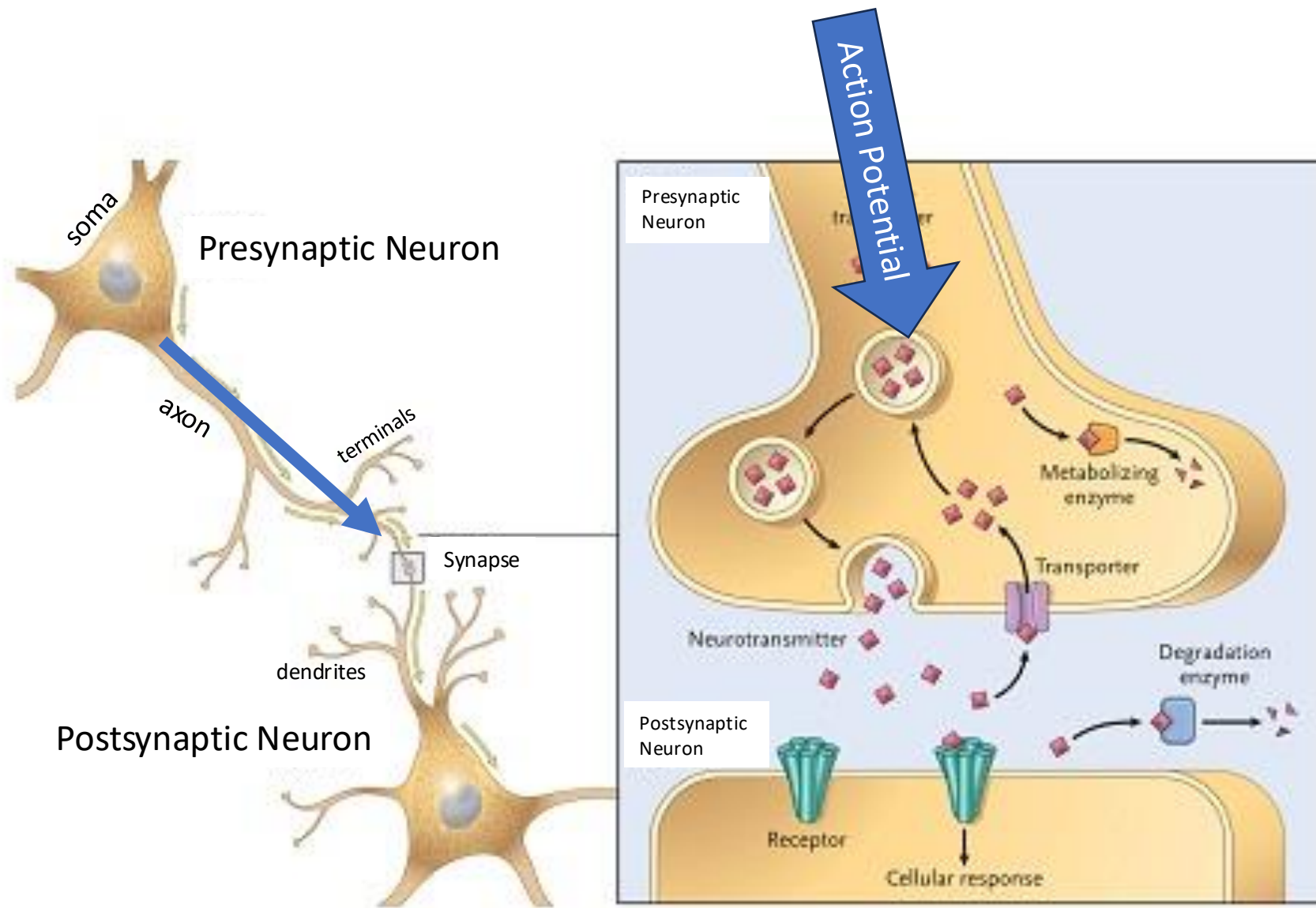
Synaptic Consolidation

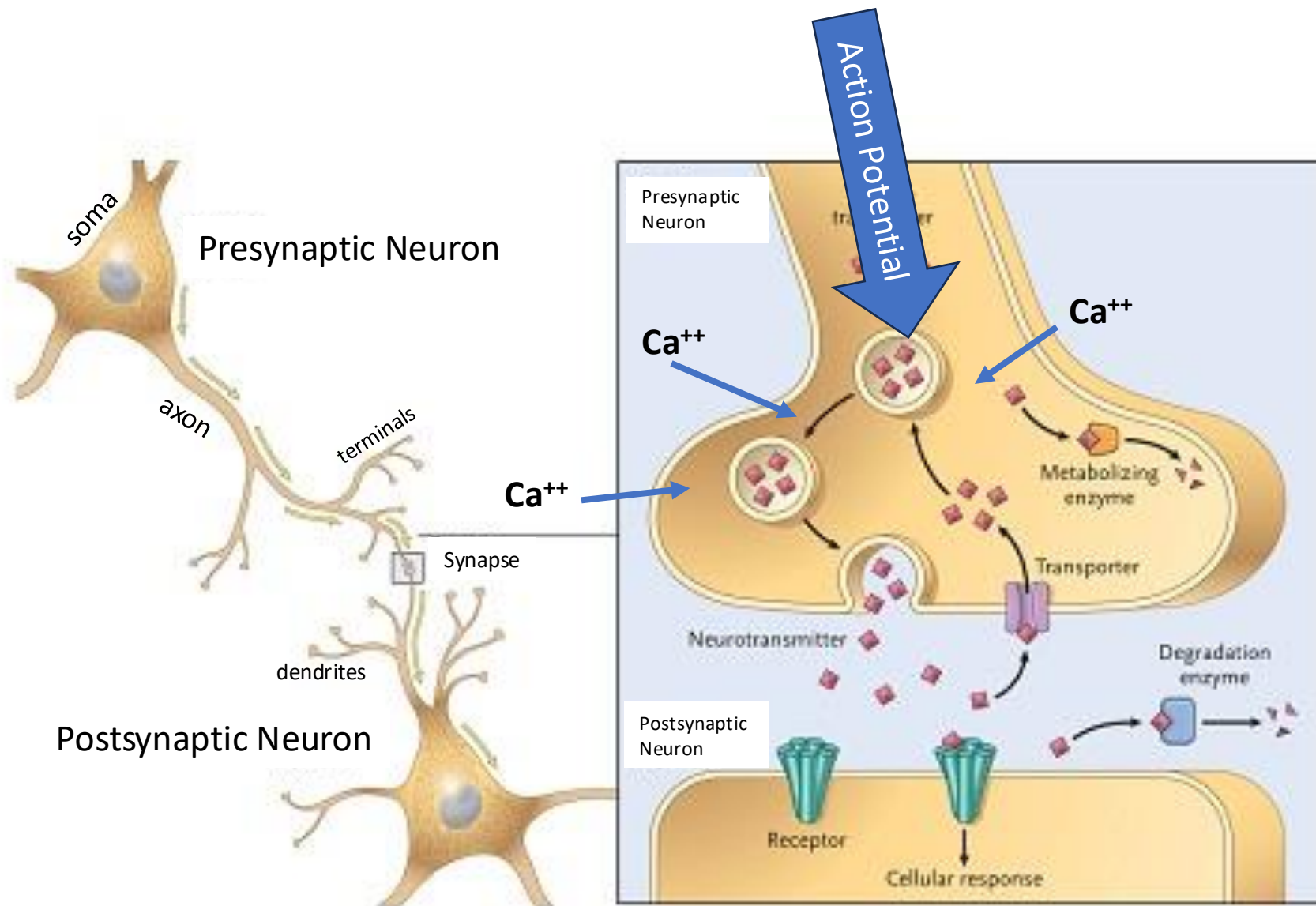
Basics of the Synapse

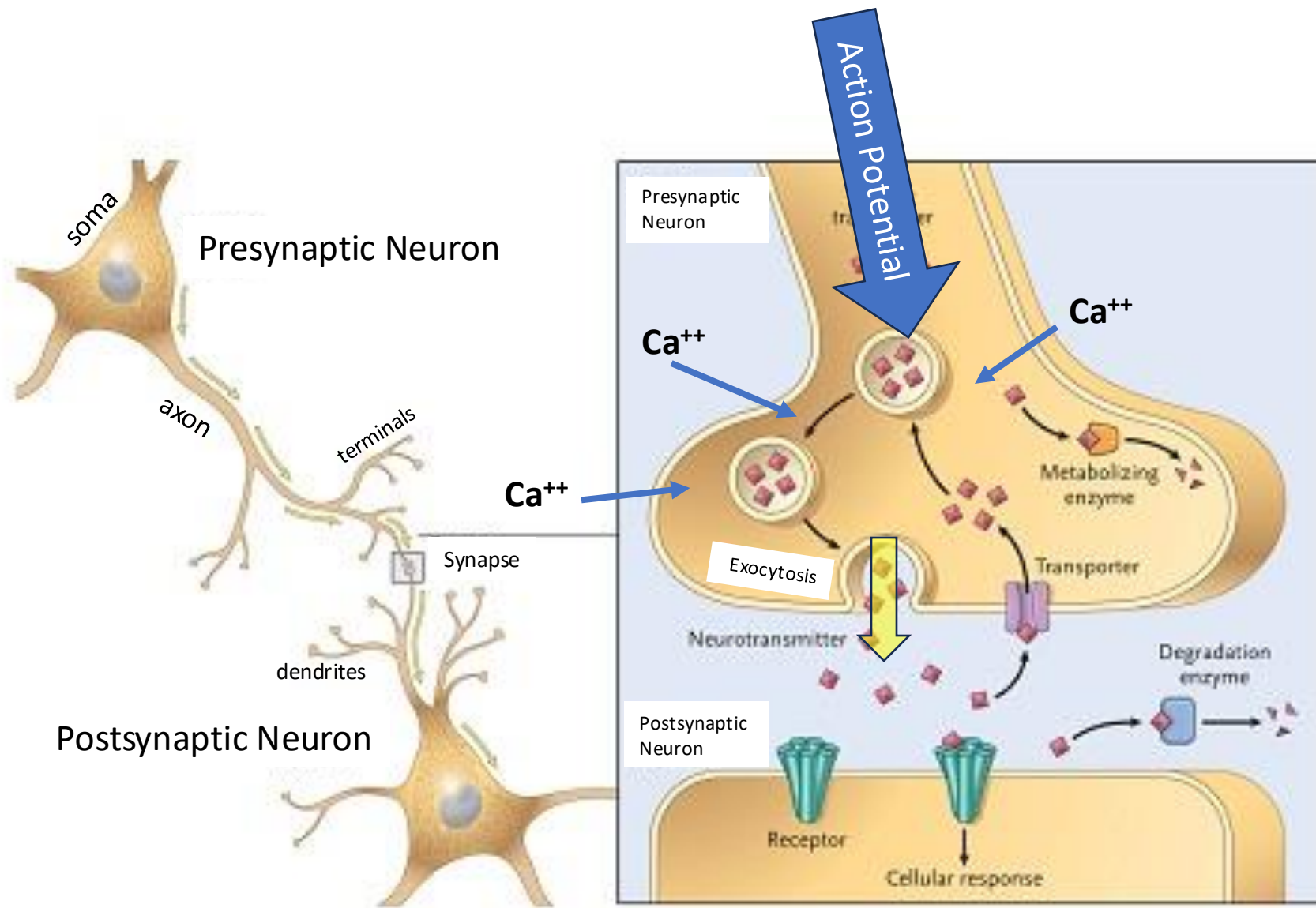
“Learning and memory require the formation of new neural networks in the brain. A key mechanism underlying this process is synaptic plasticity at excitatory synapses, which connect neurons into networks.” (Kennedy, 2016)

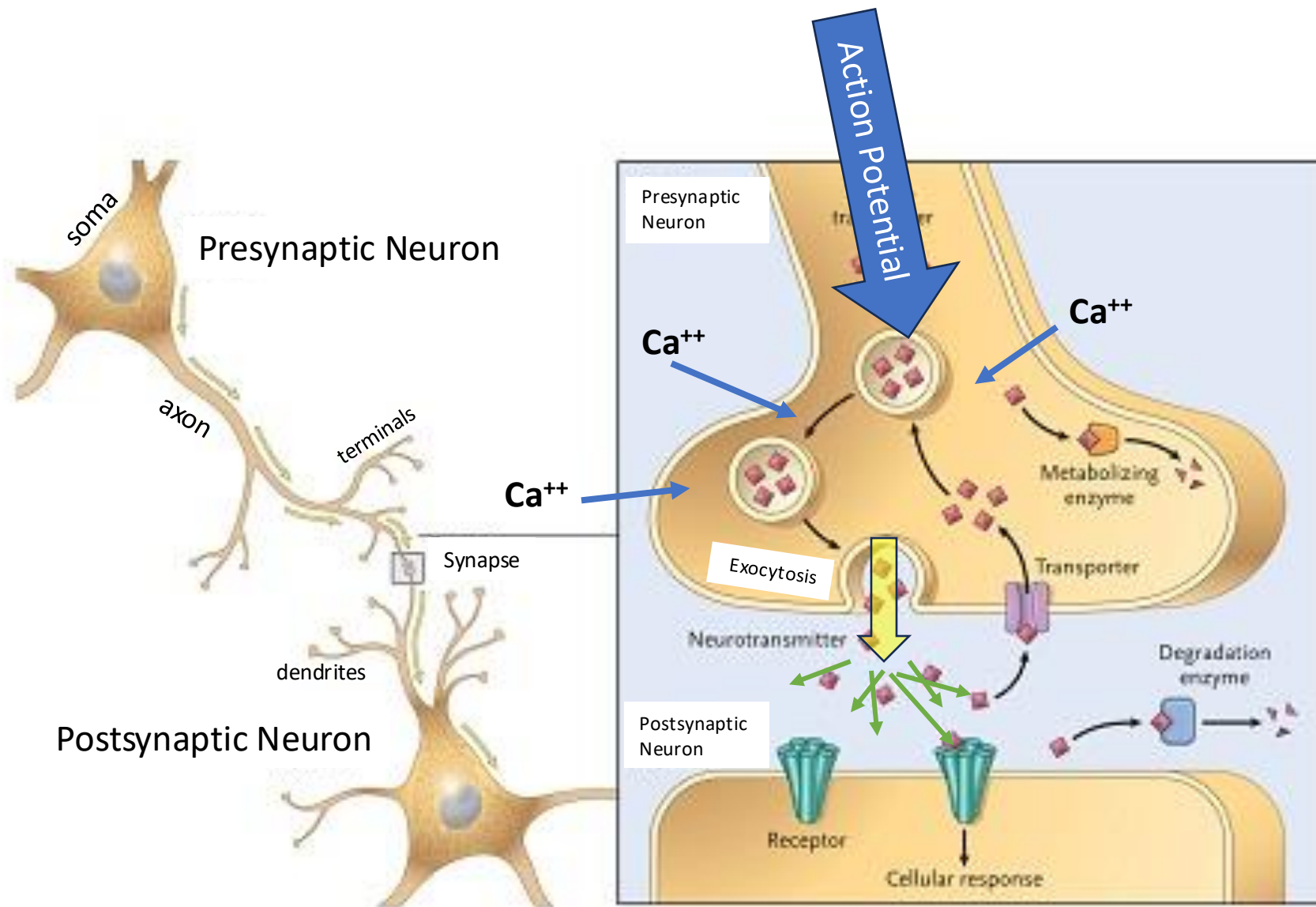
- ~86 billion neurons
- Majority are excitatory
- Most have 1000s of input synapses and 1000s of output synapses

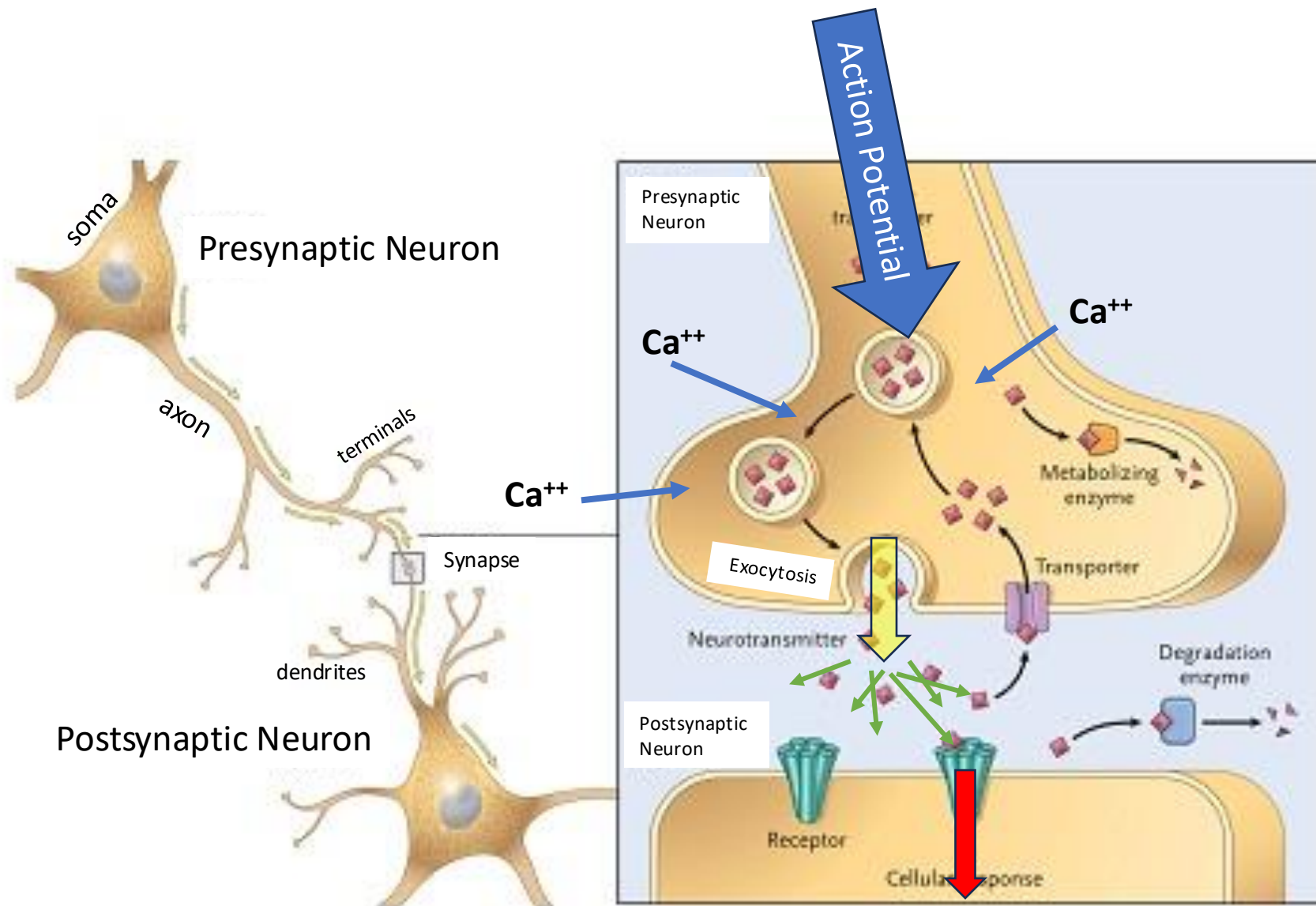




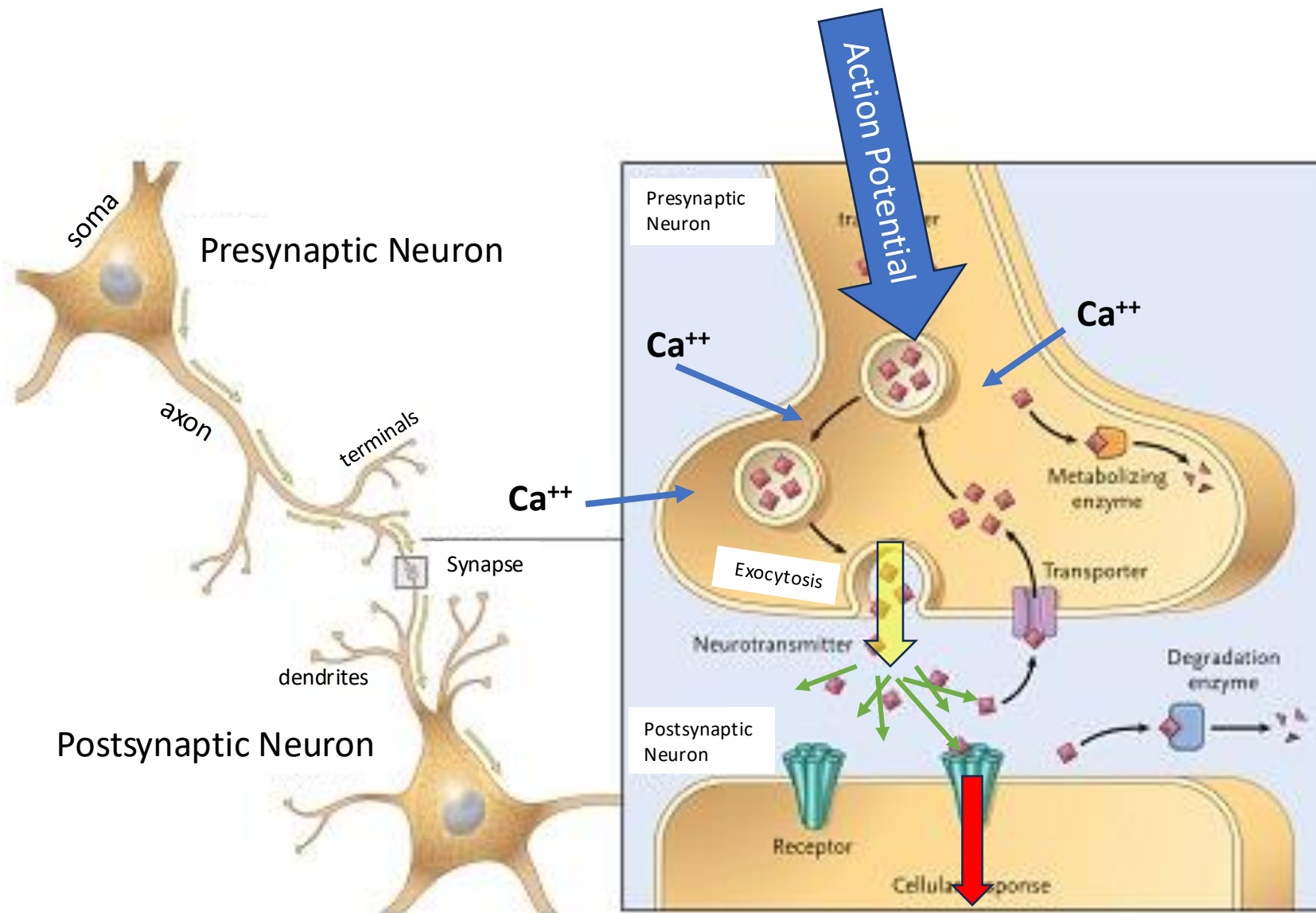








Excitatory or Inhibitory Response

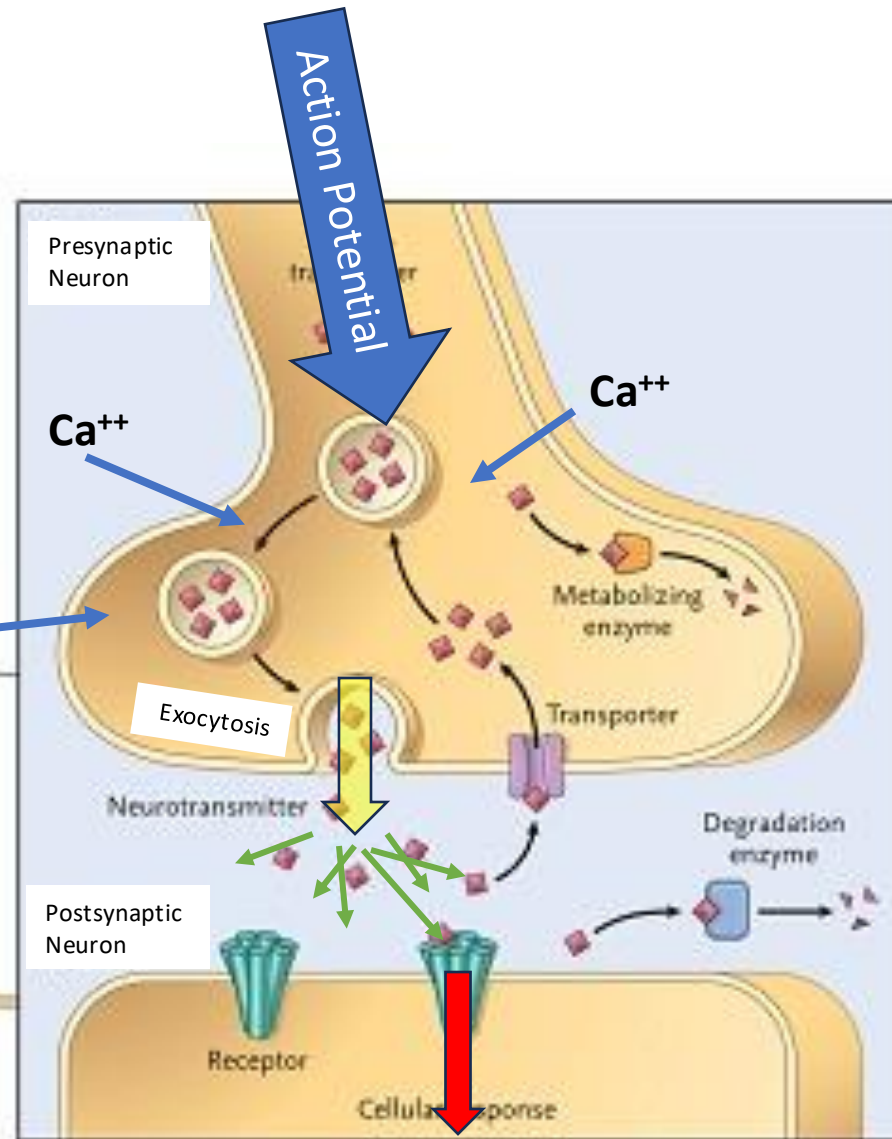
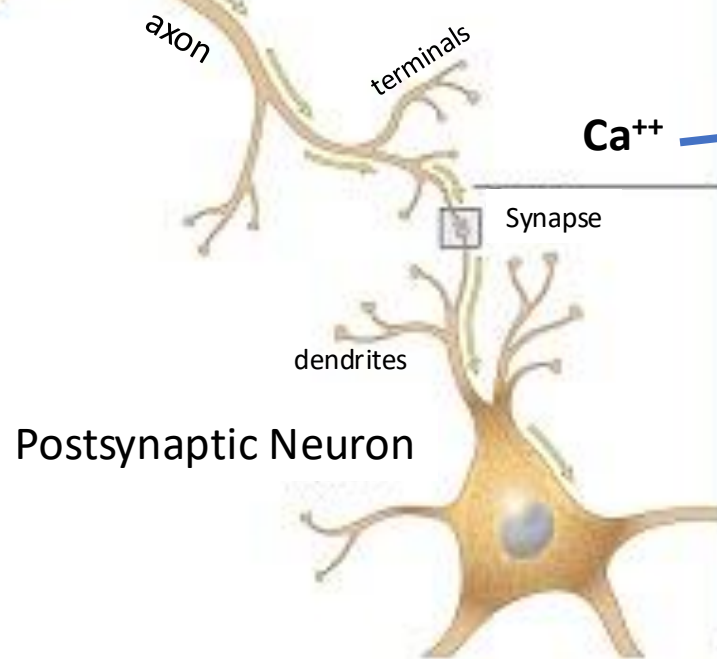


Excitatory or Inhibitory Response

How to make this postsynaptic neuron fire:

- This excitatory presynaptic neuron fires repeatedly and quickly
- Many excitatory presynaptic neurons all around it fire around the same time

Info is “stored” in this synapse (i.e., *learning* has taken place) when there is an increase/decrease in the strength of the presynaptic neuron’s effect on the postsynaptic neuron



Excitatory or Inhibitory Response

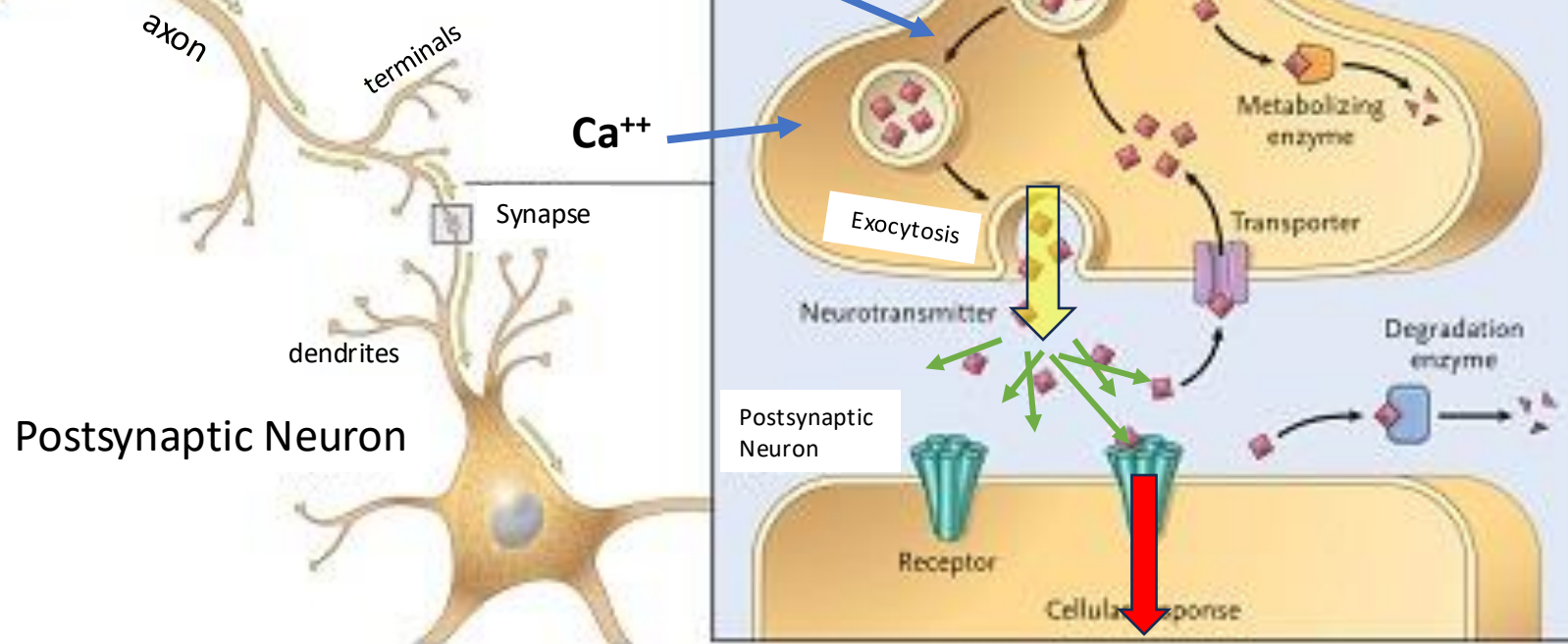
How to make this postsynaptic neuron fire:

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Effect of Presynaptic on Postsynaptic will:

increase with each instance that **Pre and Post are co-active**

decrease with each instance that **Pre is active** but **Post is not**, or vice versa
– Hebbian Plasticity



Excitatory or Inhibitory Response

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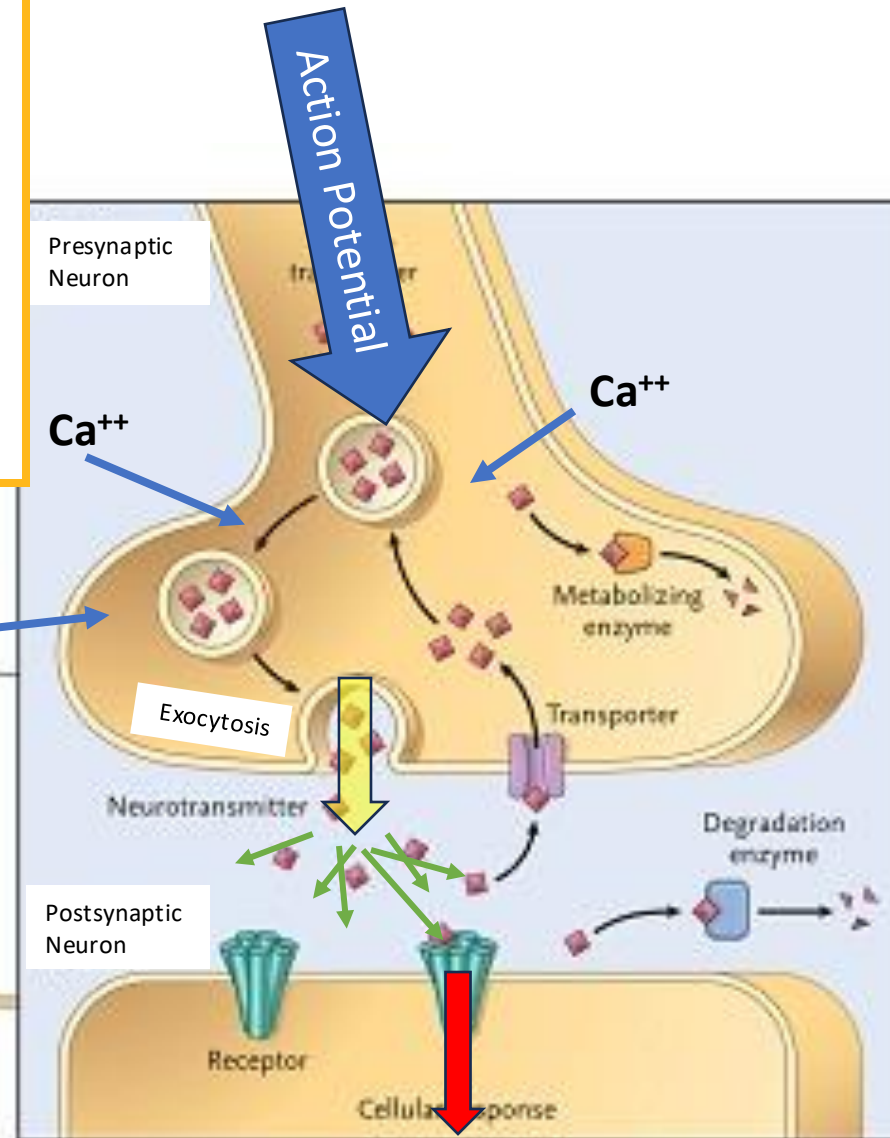
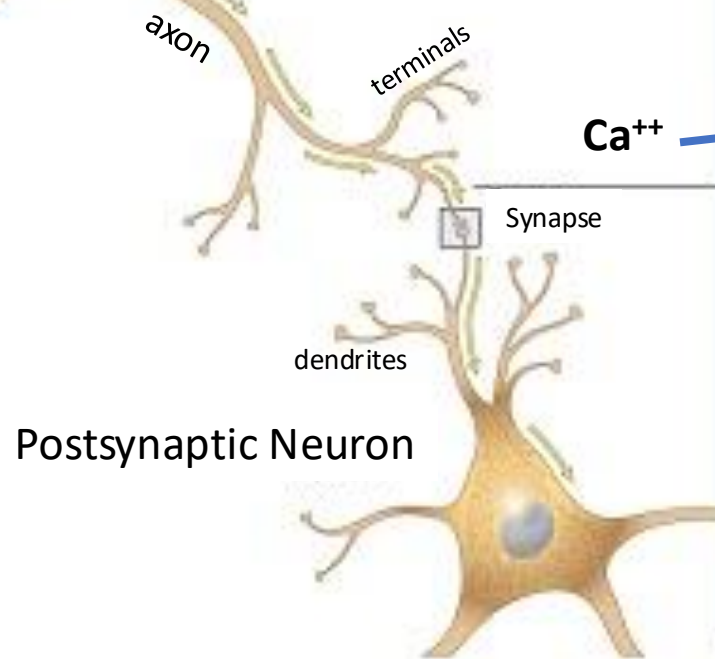
increase with each instance that **Pre and Post are co-active**

Long-Term Potentiation (LTP)

decrease with each instance that **one is active** but **the other is not**

Long-Term Depression (LTD)

– Hebbian Plasticity



Excitatory or Inhibitory Response

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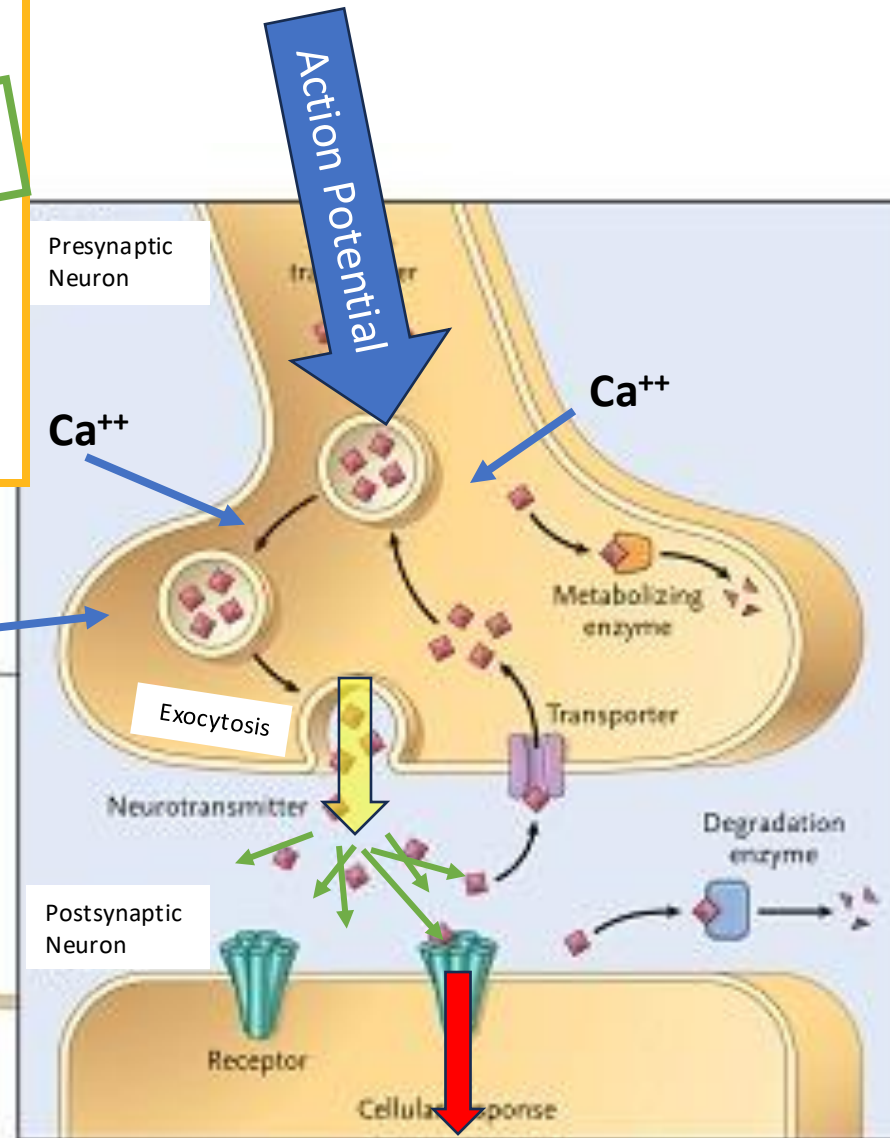
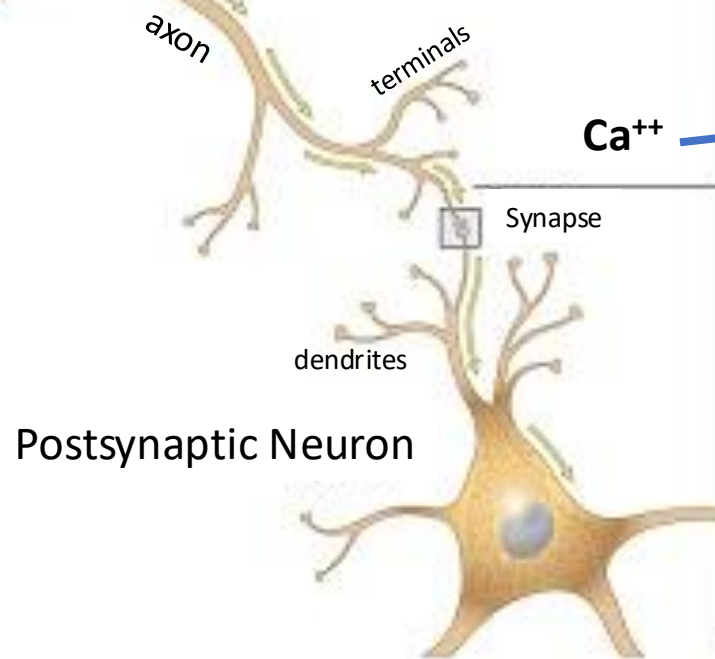
increase with each instance that Pre and Post are active

Long-Term Potentiation (LTP)

ONLY OCCURS IN **GLUTAMATERGIC SYNAPSES** (maybe)

one is active but the other is not
Long-Term Depression (LTD)

– Hebbian Plasticity



Excitatory or Inhibitory Response

How to make this postsynaptic neuron fire:

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- Many excitatory presynaptic neurons all around it fire around the same time

Glutamate & Synaptic Consolidation

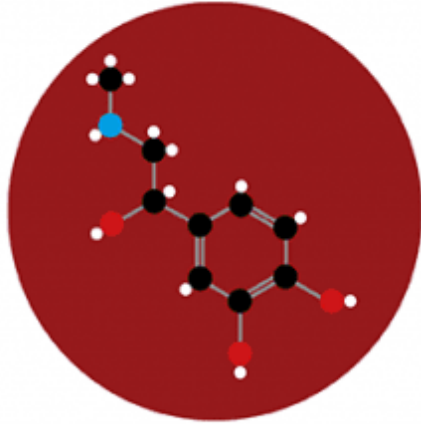
“Learning and memory require the formation of new neural networks in the brain. A key mechanism underlying this process is synaptic plasticity at excitatory synapses, which connect neurons into networks.” (Kennedy, 2016)

“Excitatory synaptic transmission happens when **glutamate**, the excitatory neurotransmitter, activates receptors on the postsynaptic neuron. Synaptic plasticity is a higher-level process in which the strength of excitatory synapses is altered in response to the pattern of activity at the synapse.” (Kennedy, 2016)

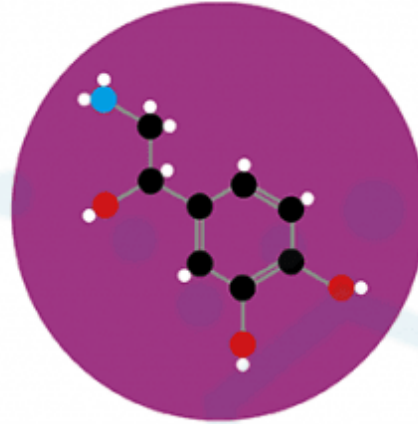
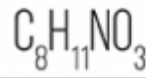
ONE PER NEURON

ESSENTIAL NEUROTRANSMITTERS

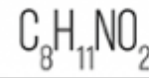
ADRENALINE
THE "GHT OR FLIGHT" NEUROTRANSMITTER



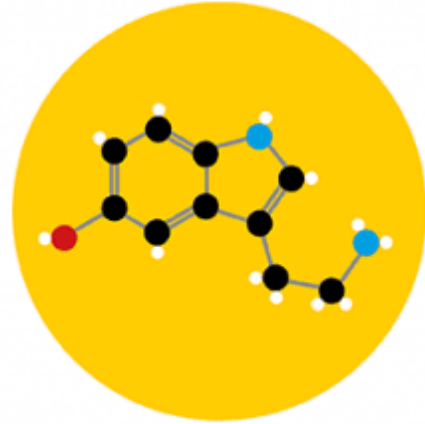
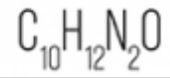
NORADRENALINE
THE CONCENTRATION NEUROTRANSMITTER



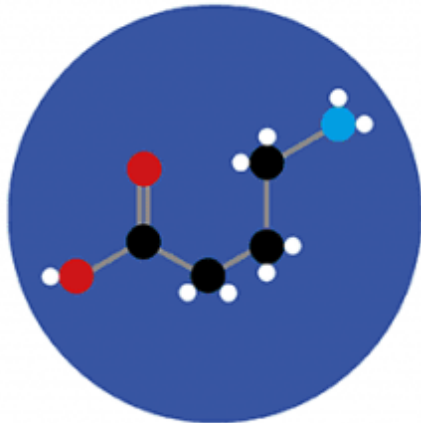
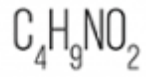
DOPAMINE
THE PLEASURE NEUROTRANSMITTER



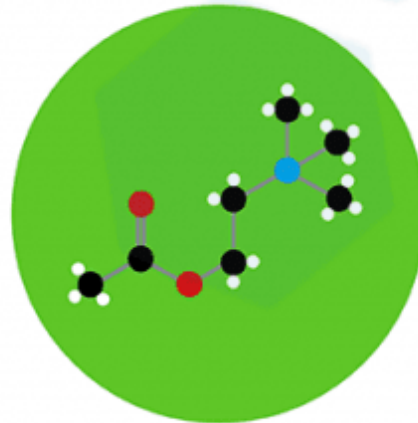
SEROTONIN
THE MOOD NEUROTRANSMITTER



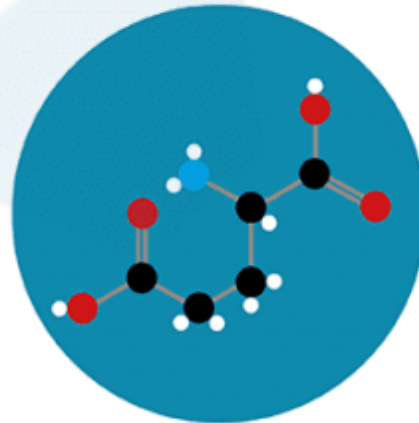
γ -AMINO BUTYRIC ACID
THE CALMING NEUROTRANSMITTER



ACETYLCHOLINE
THE LEARNING NEUROTRANSMITTER

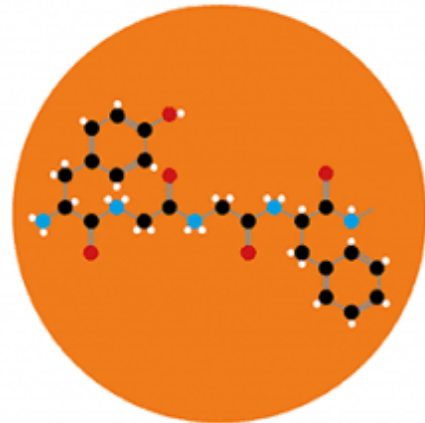


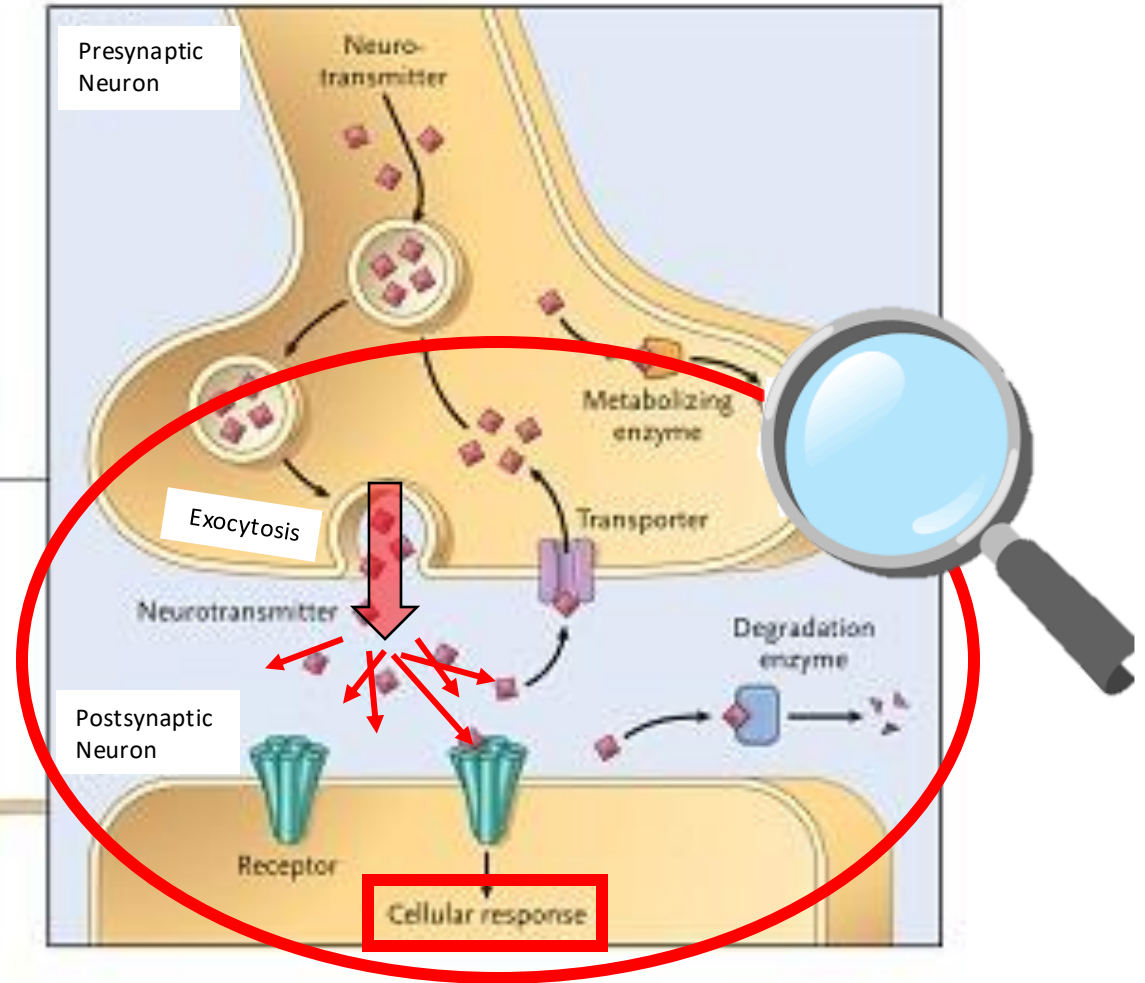
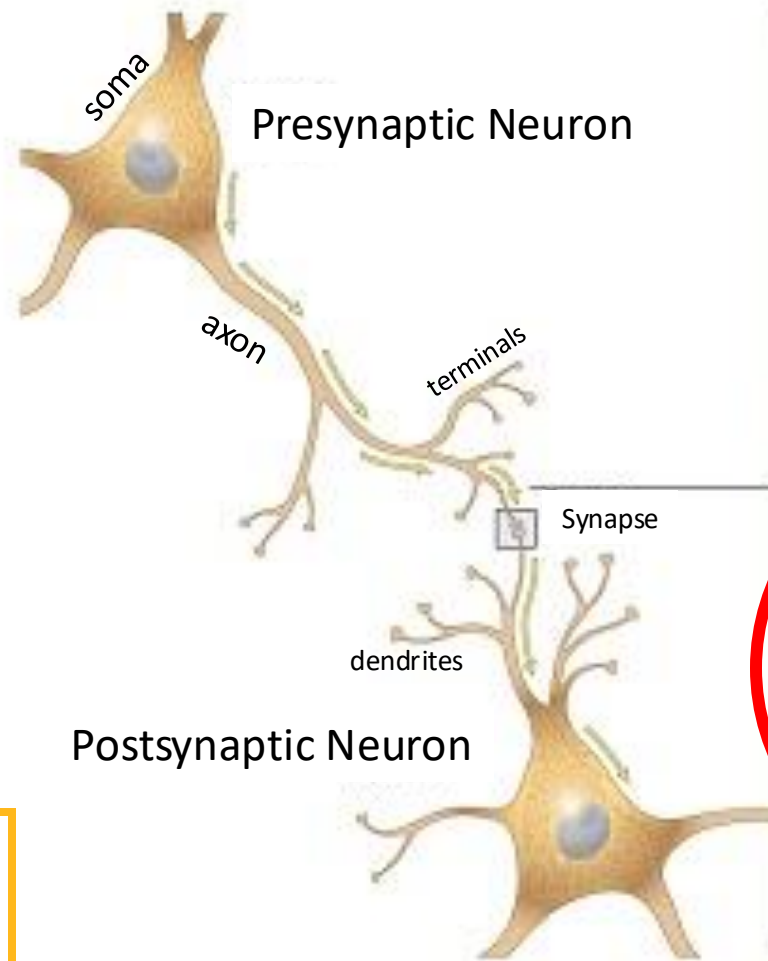
GLUTAMATE
THE MEMORY NEUROTRANSMITTER



ENDORPHINS
THE EUPHORIA NEUROTRANSMITTERS

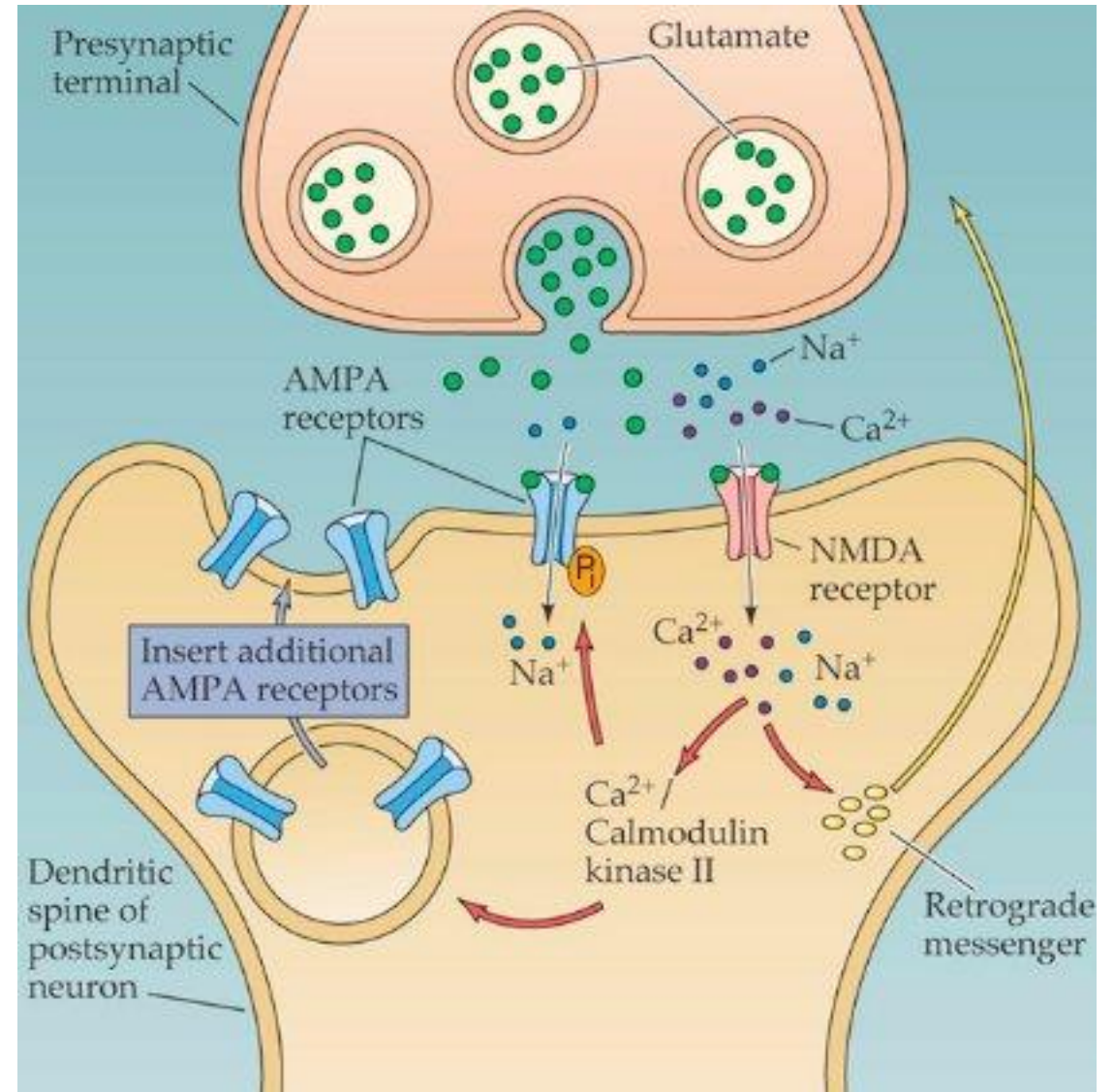
20+ TYPES IN THE HUMAN BODY





Let's zoom in on what's happening at Glutamatergic synapses right here...

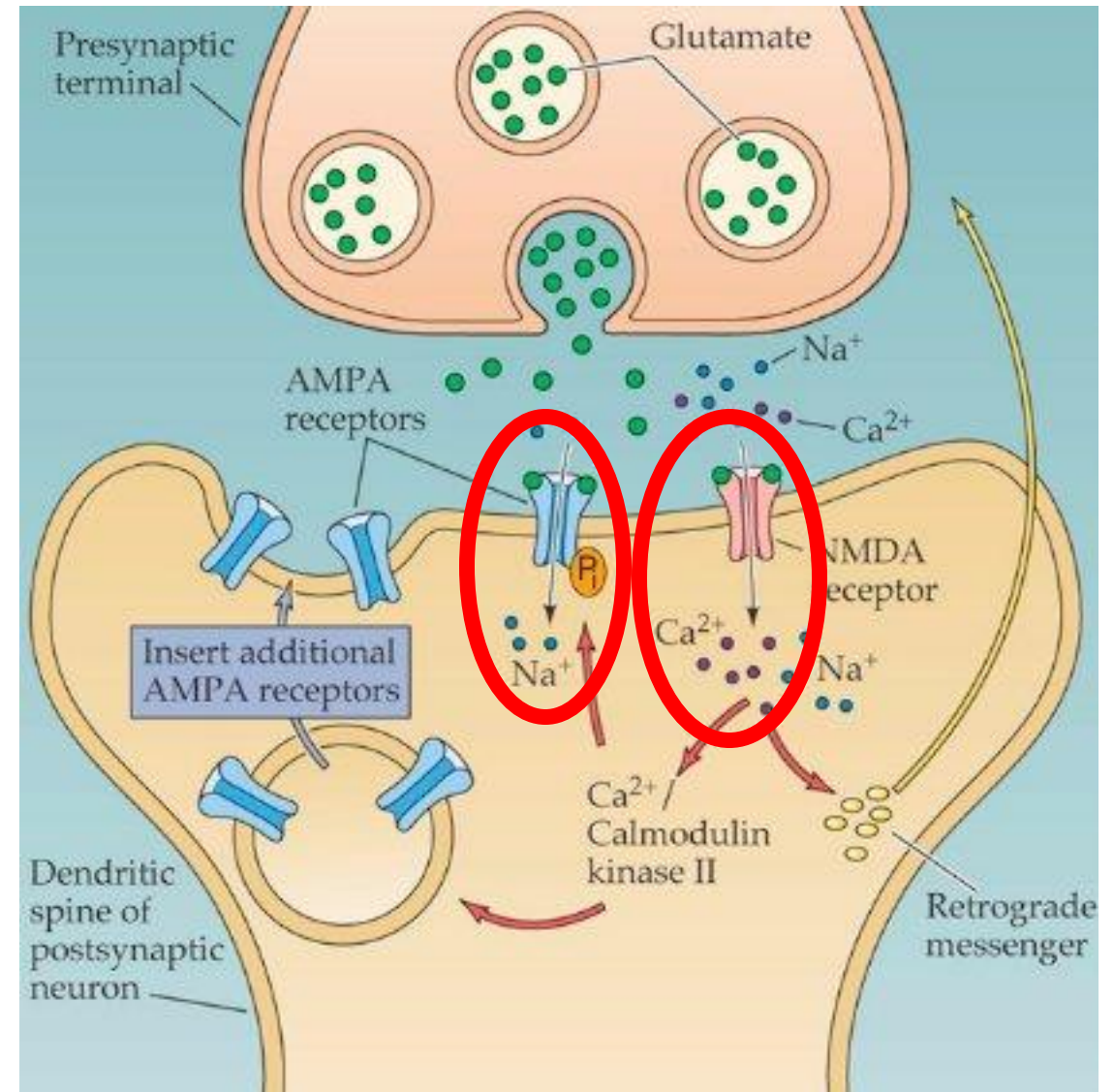
Typical glutamatergic
synapse



Glutamate Receptor Types

AMPA

NMDA

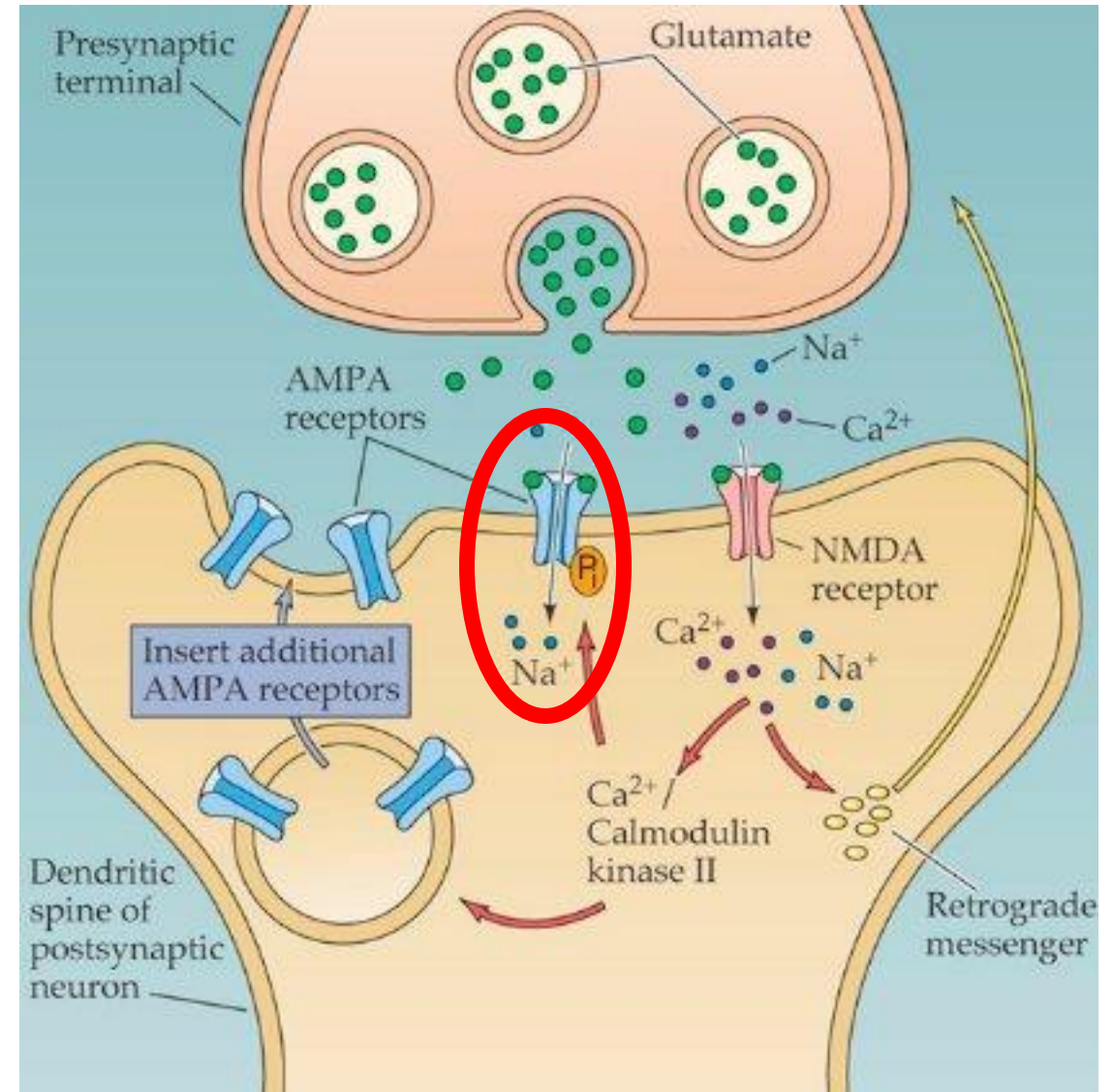


Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it

NMDA

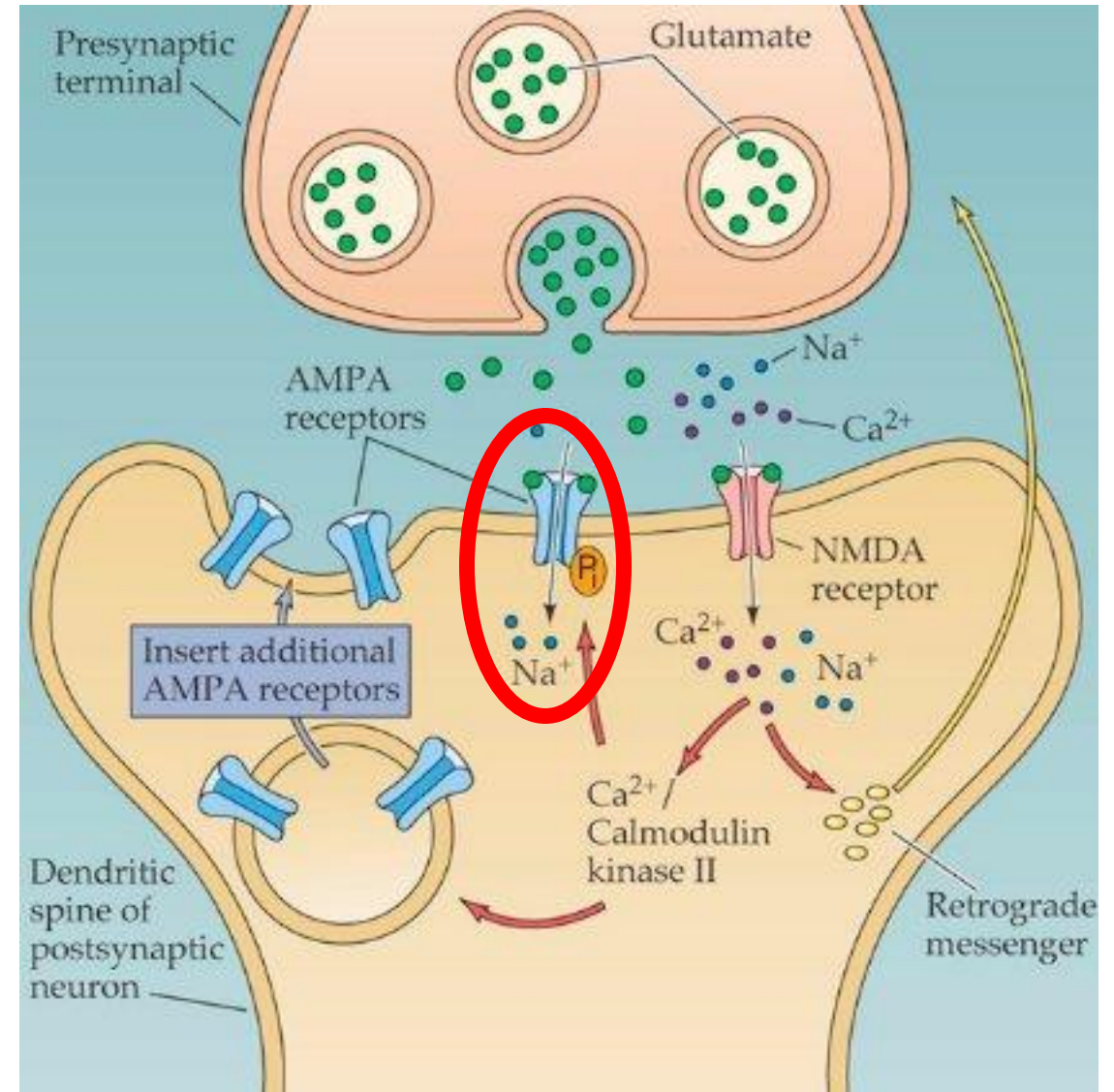


Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow Na^+ in

NMDA

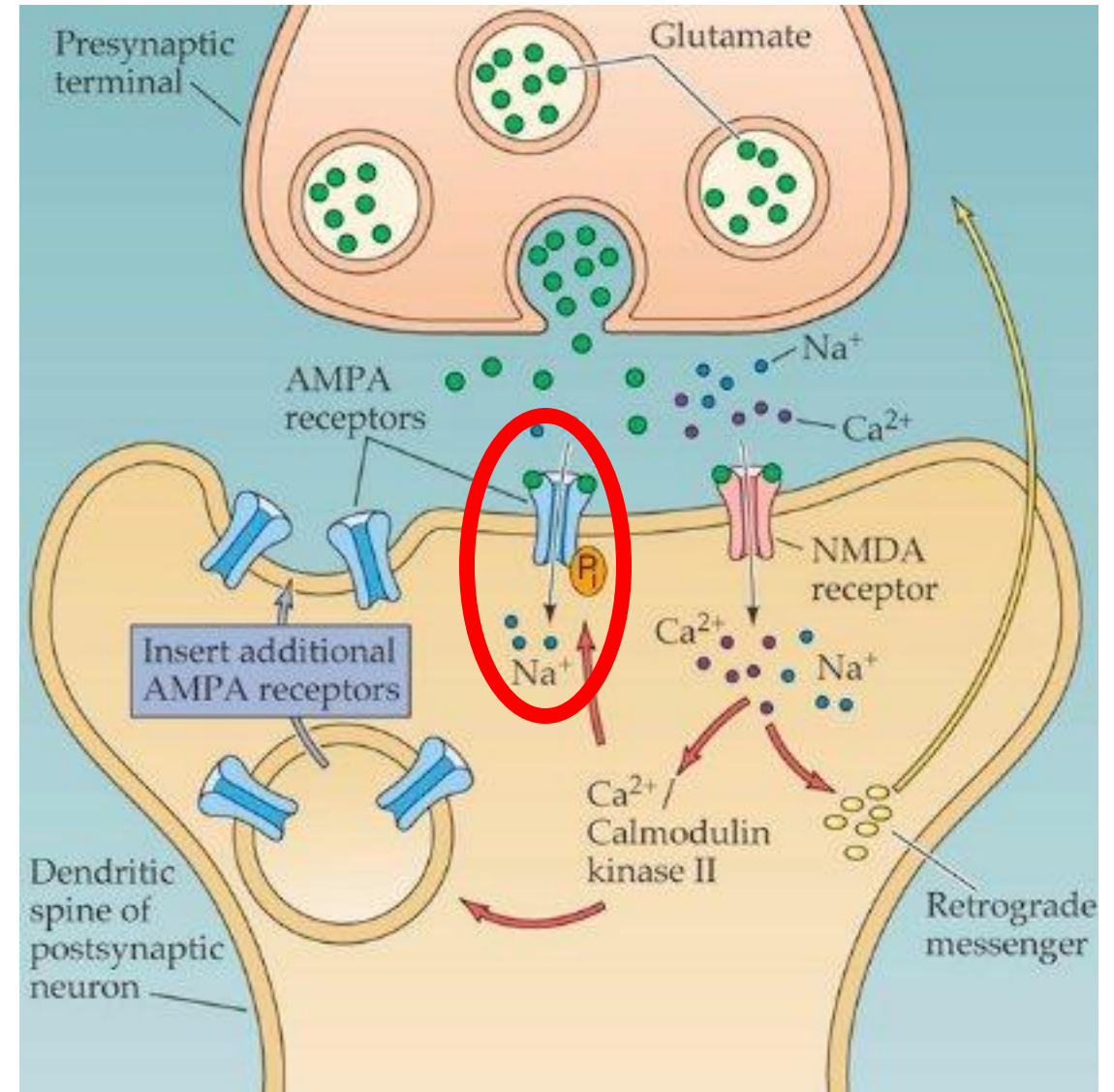


Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow Na^+ in
- Depolarizes (*excites*) postsynaptic neuron

NMDA



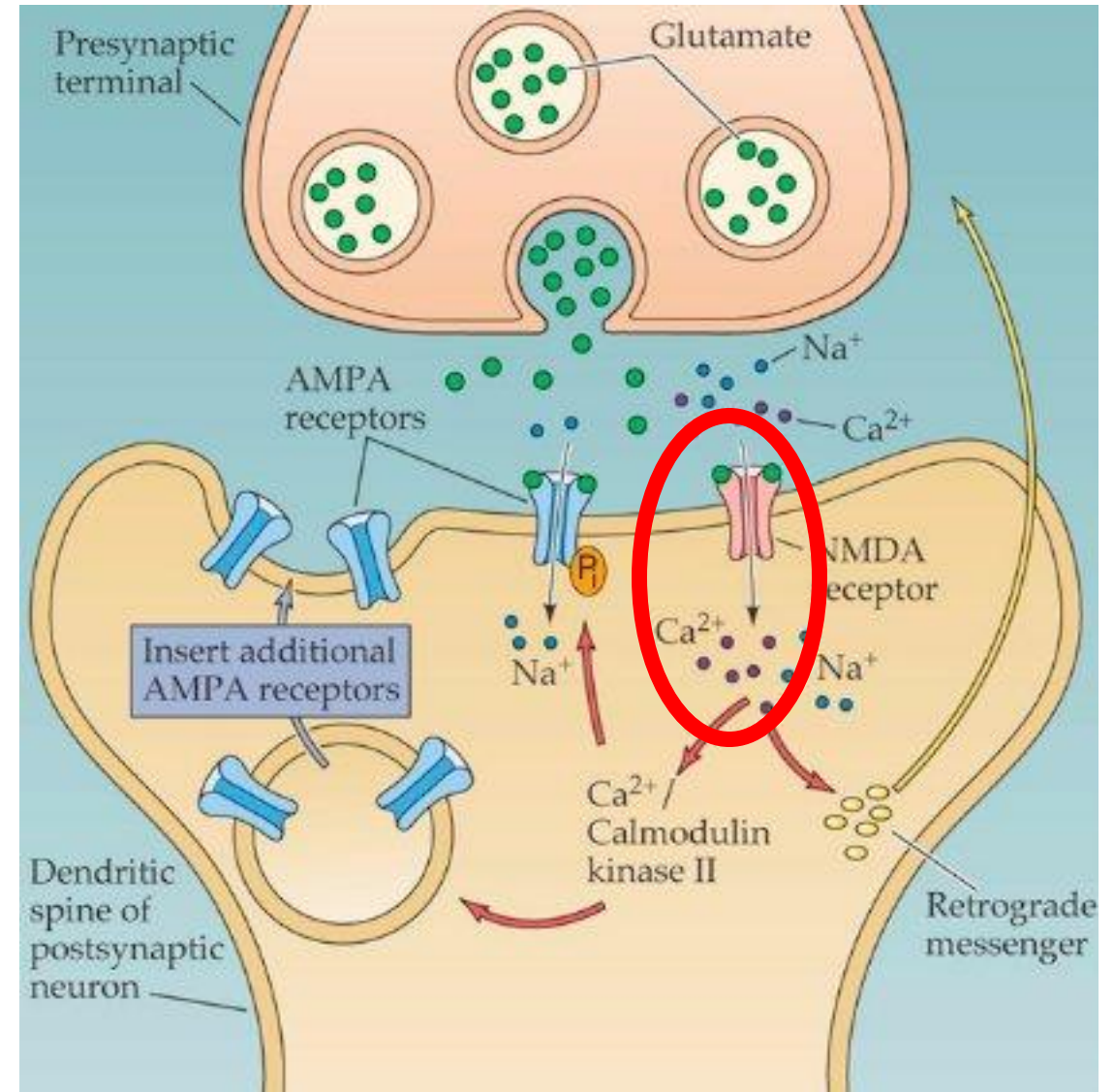
Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow Na^+ in
- Depolarizes (*excites*) postsynaptic neuron

NMDA

- Open when Glutamate from presynaptic binds to it AND postsynaptic is firing



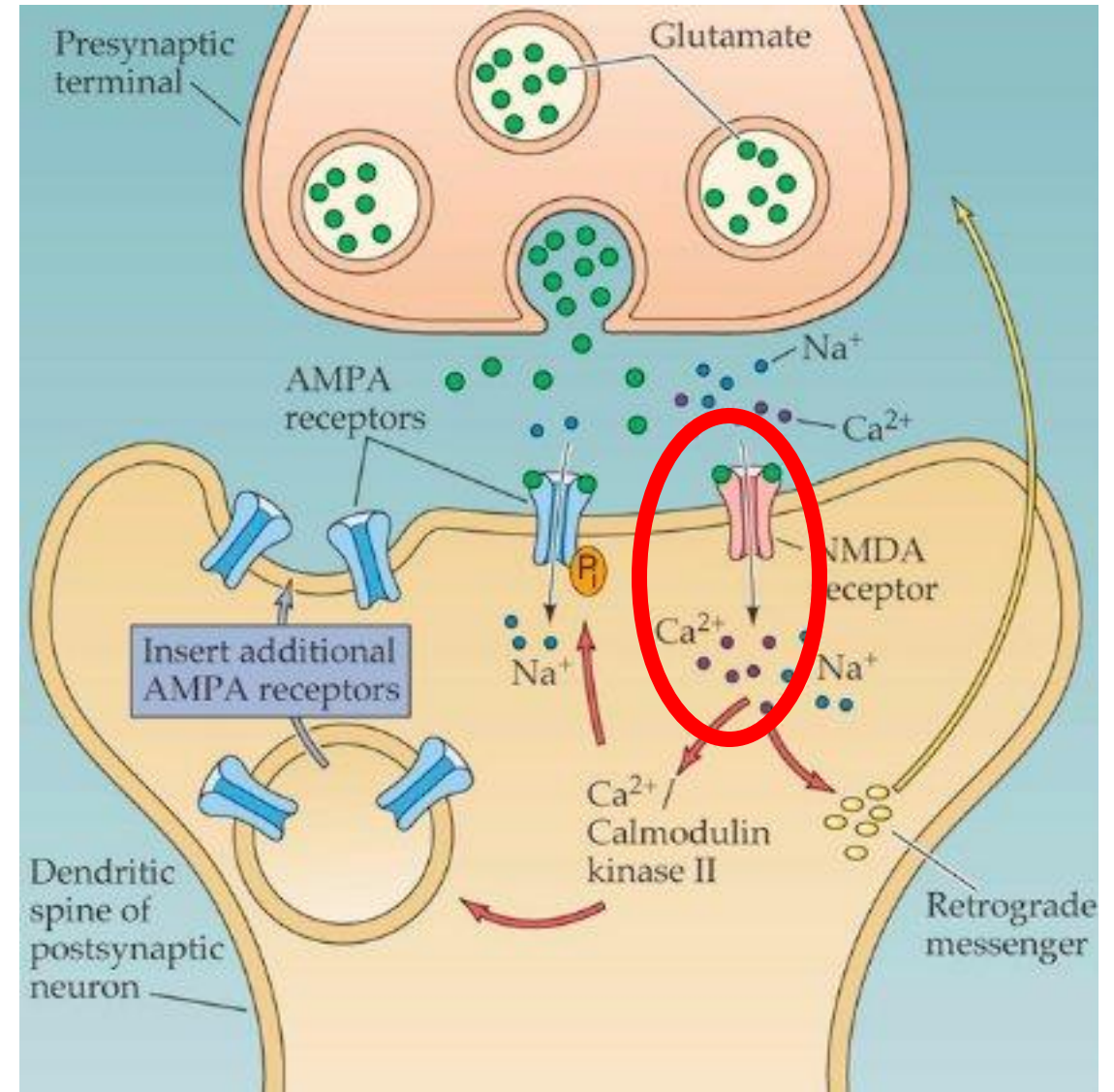
Glutamate Receptor Types

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- Open when Glutamate from presynaptic binds to it
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- Open when Glutamate from presynaptic binds to it AND postsynaptic is firing
- Allow Ca^{++} in



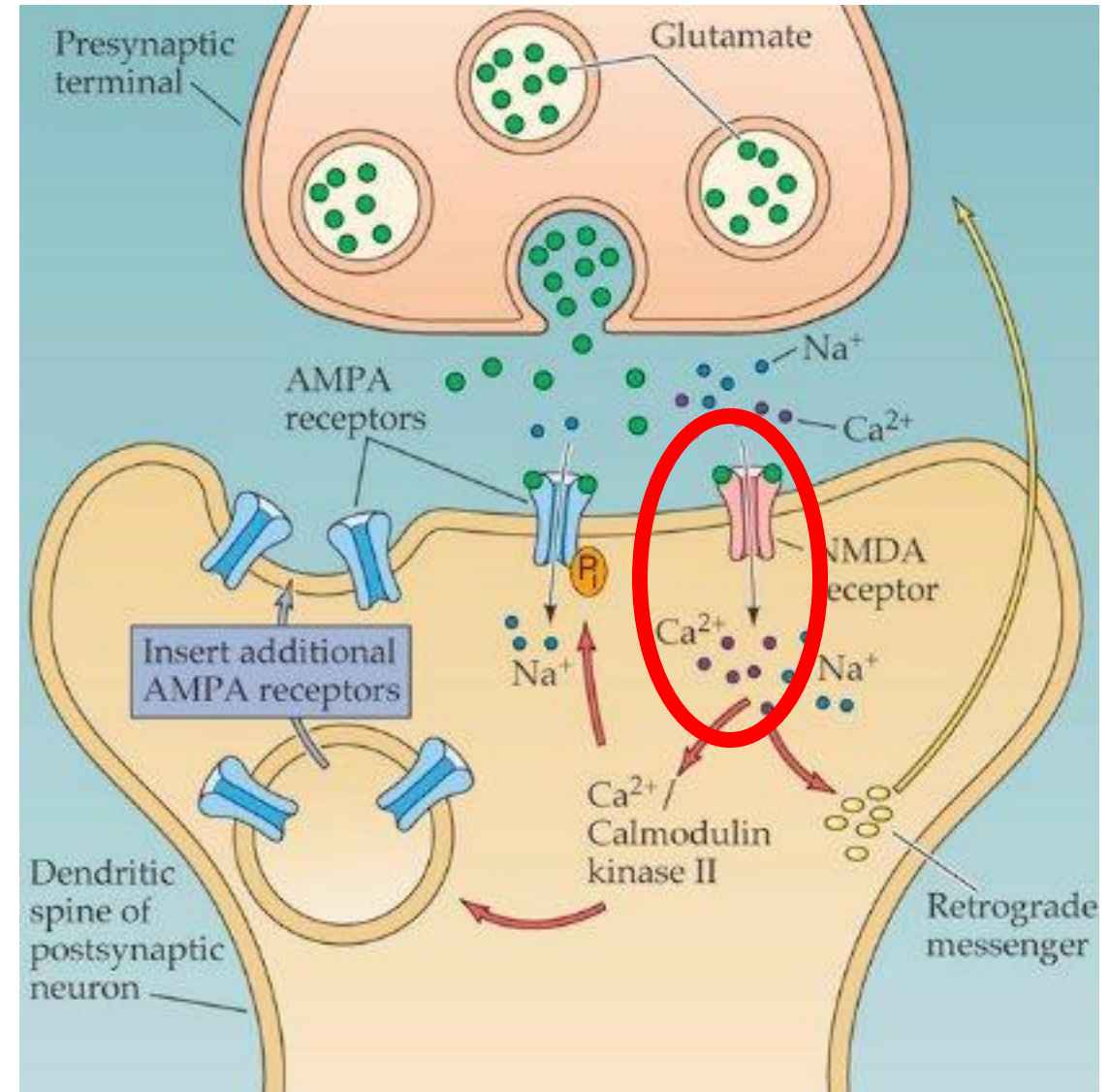
Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow Na^+ in
- Depolarizes (*excites*) postsynaptic neuron

NMDA

- Open when Glutamate from presynaptic binds to it AND postsynaptic is firing
- Allow Ca^{++} in
- Triggers adaptation of postsynaptic neuron by adding new AMPA receptors at that synapse



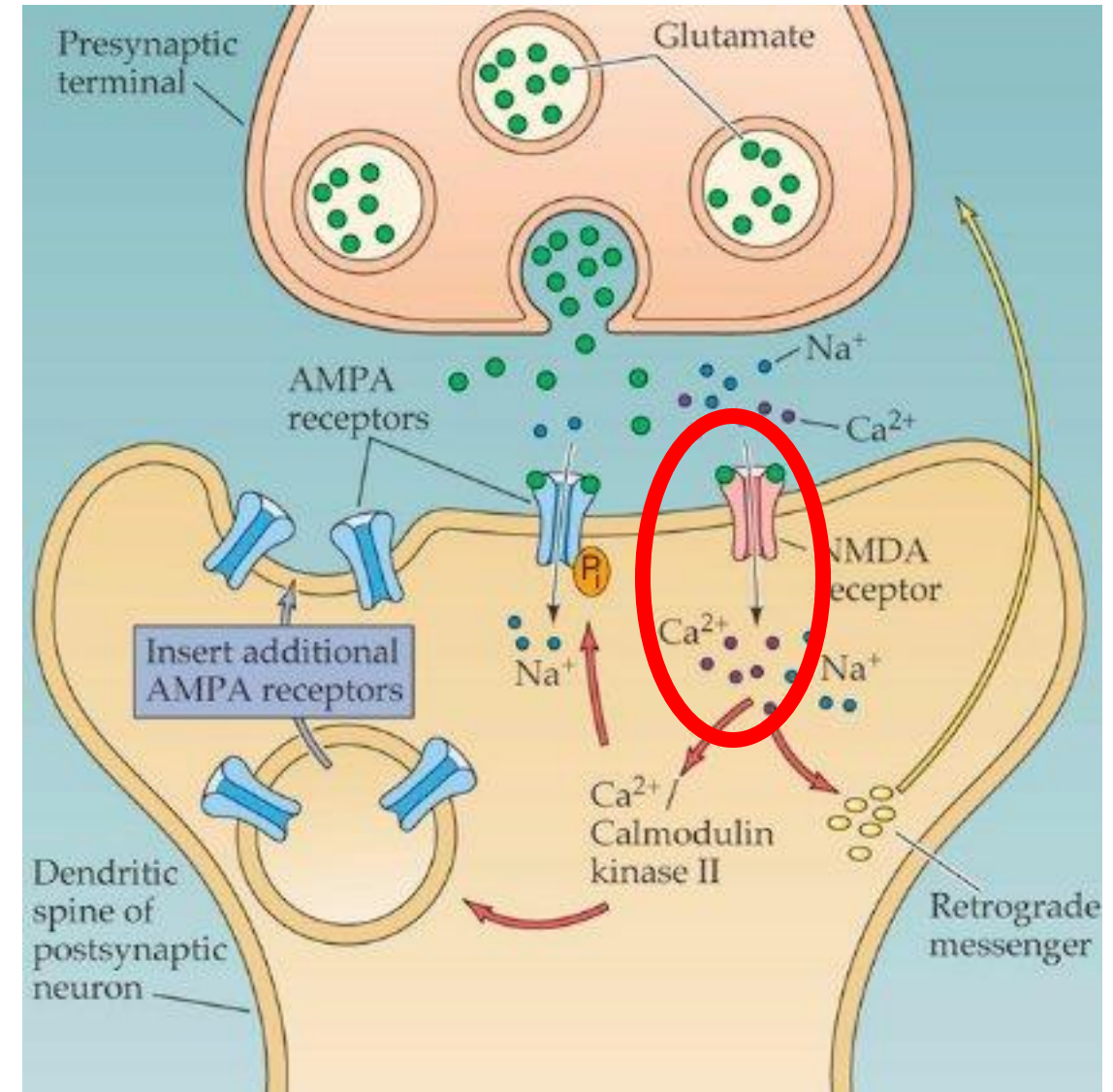
Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow **Na⁺** in
- **Depolarizes (*excites*) postsynaptic neuron**

NMDA

- Open when Glutamate from presynaptic binds to it AND postsynaptic is firing
- Allow **Ca⁺⁺** in
- **Triggers adaptation of postsynaptic neuron by adding new AMPA receptors at that synapse**



Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow Na^+ in
- **Depolarizes (*excites*) postsynaptic neuron**

NMDA

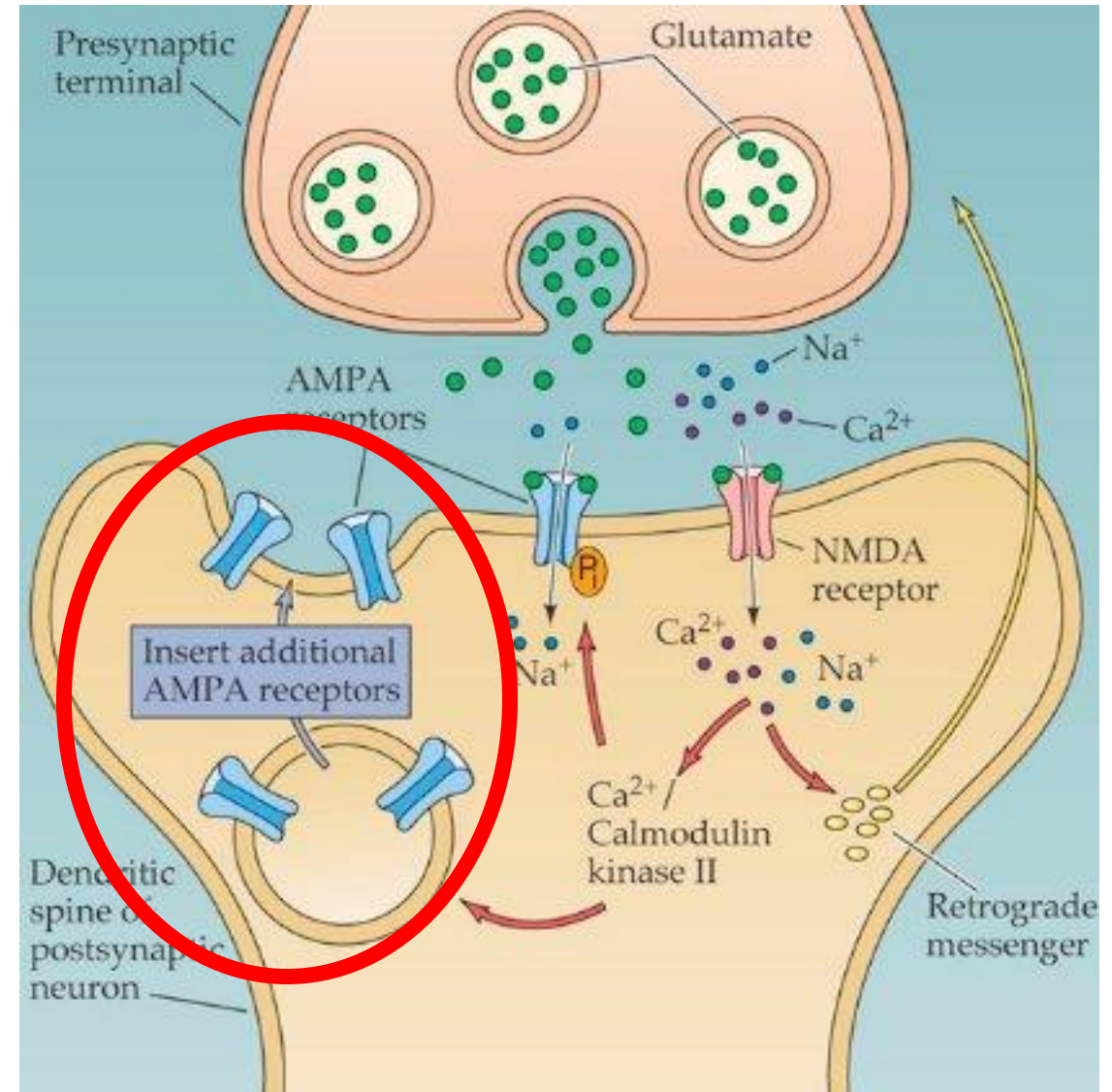
Presynaptic releases Glutamate that binds to receptor AND postsynaptic neuron fires action potential



Receptor opens causing Calcium to enter postsynaptic



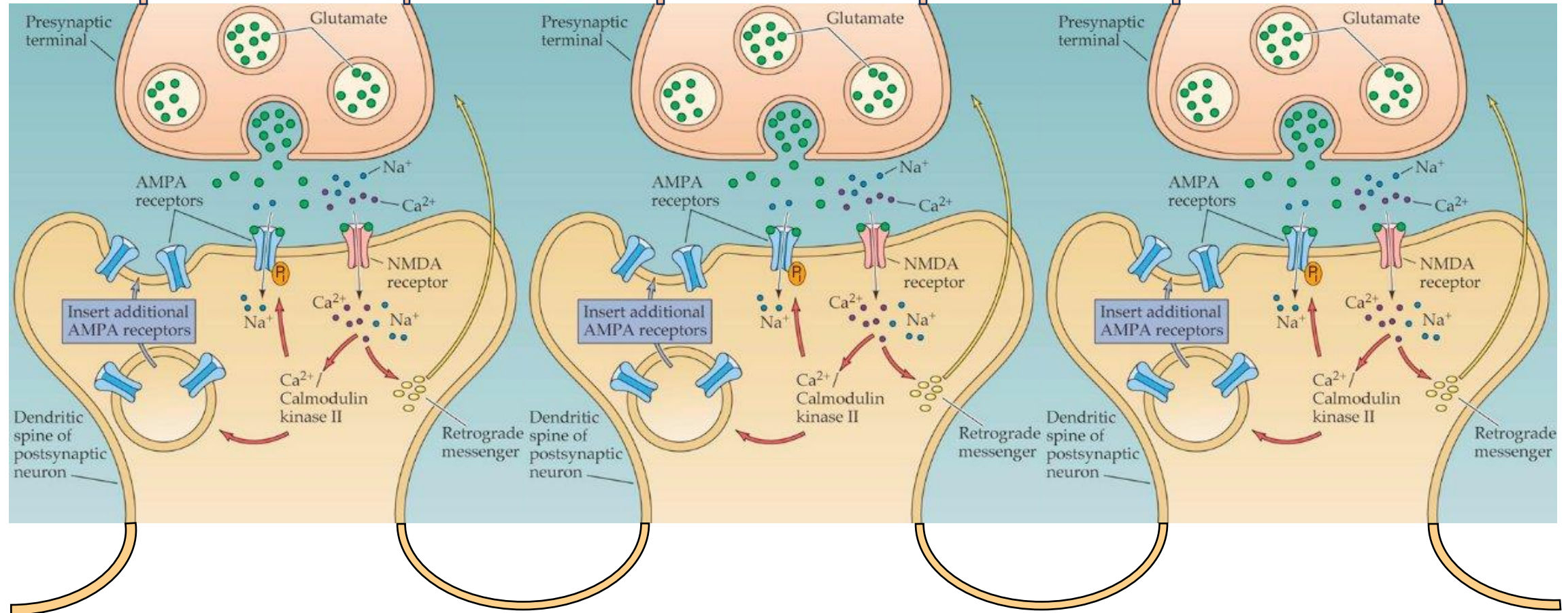
Calcium triggers insertion of new AMPA into postsynaptic



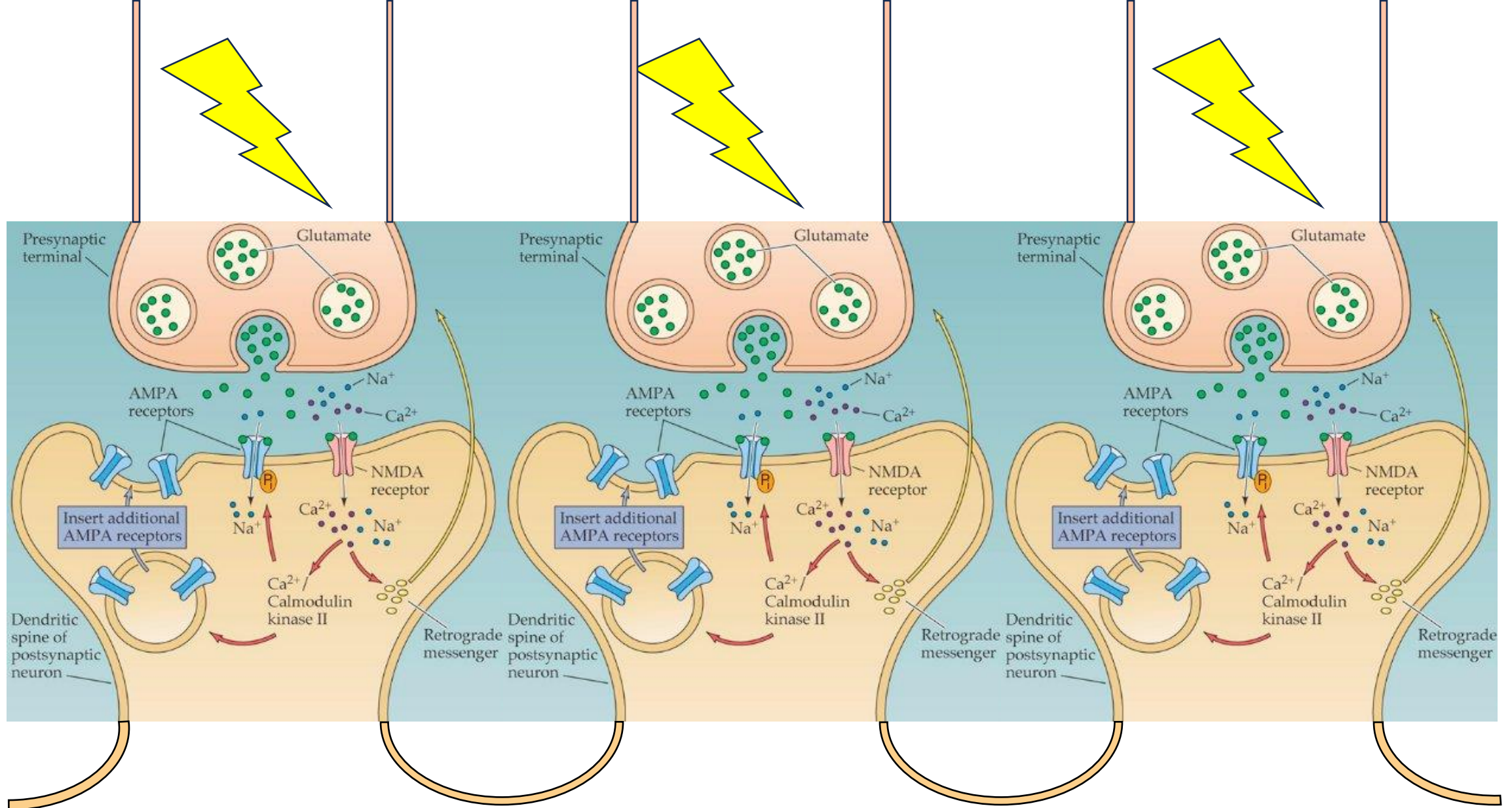
Presyn. Neuron 1

Presyn. Neuron 2

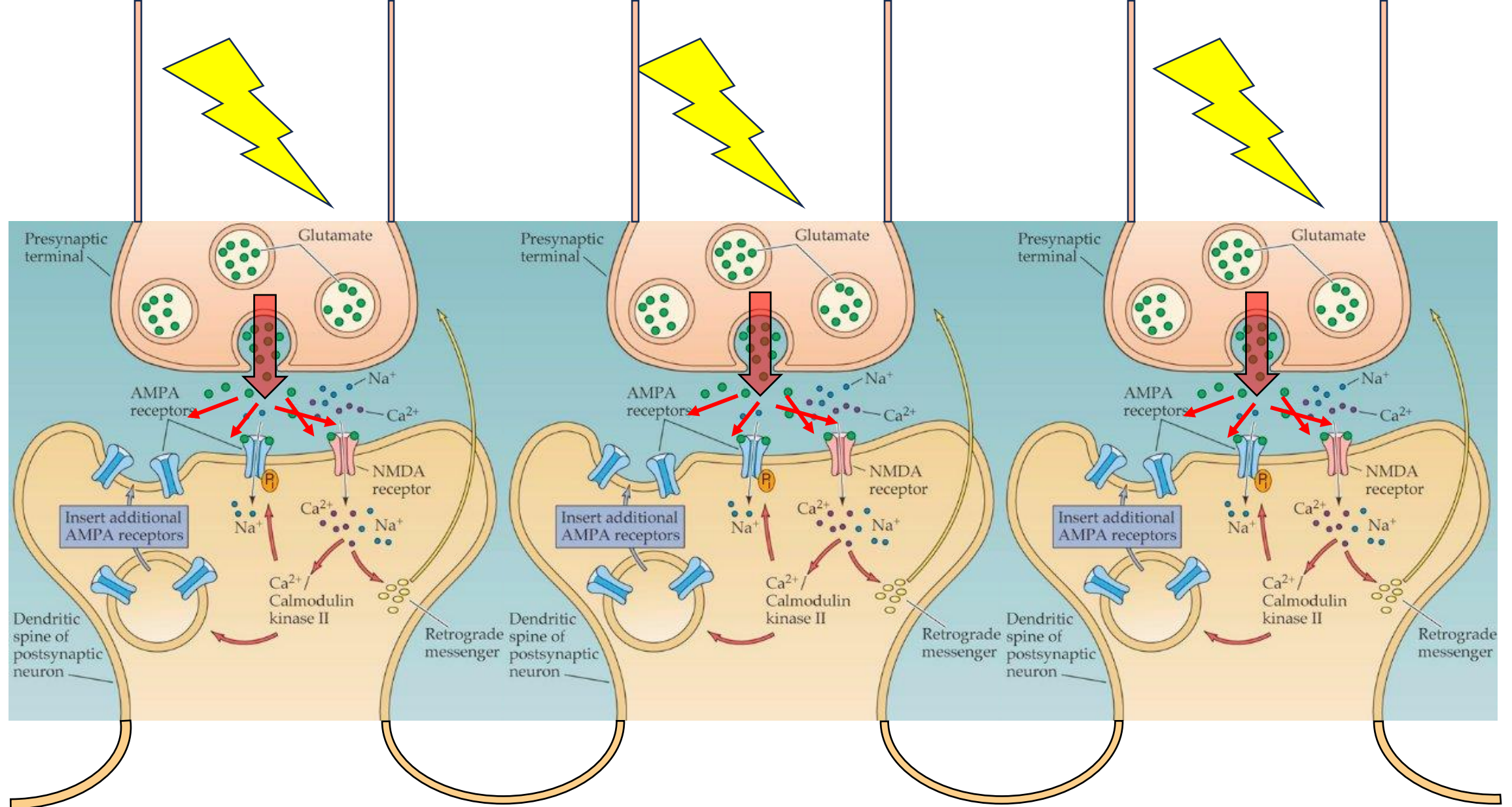
Presyn. Neuron 3



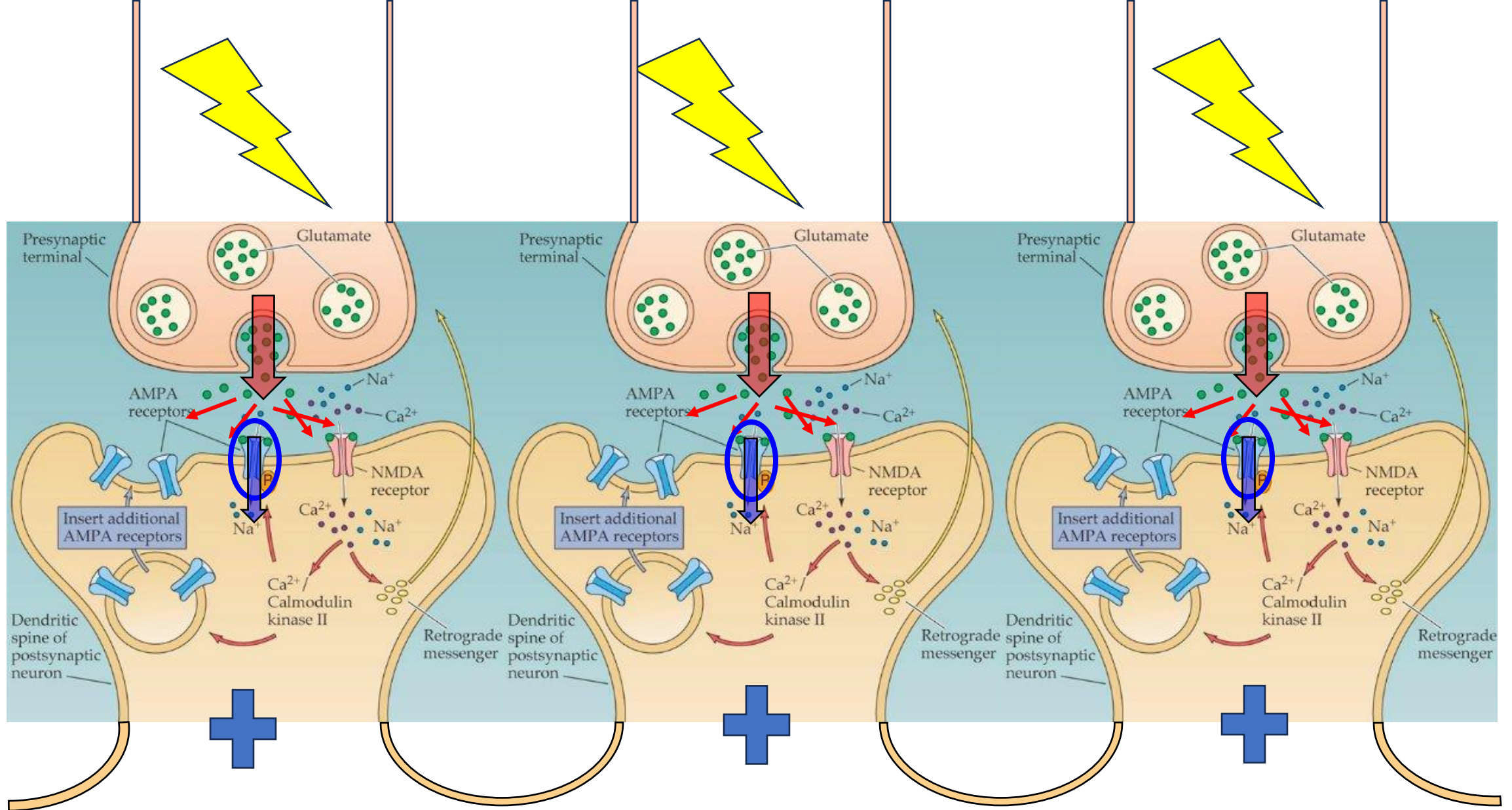
Postsynaptic Neuron



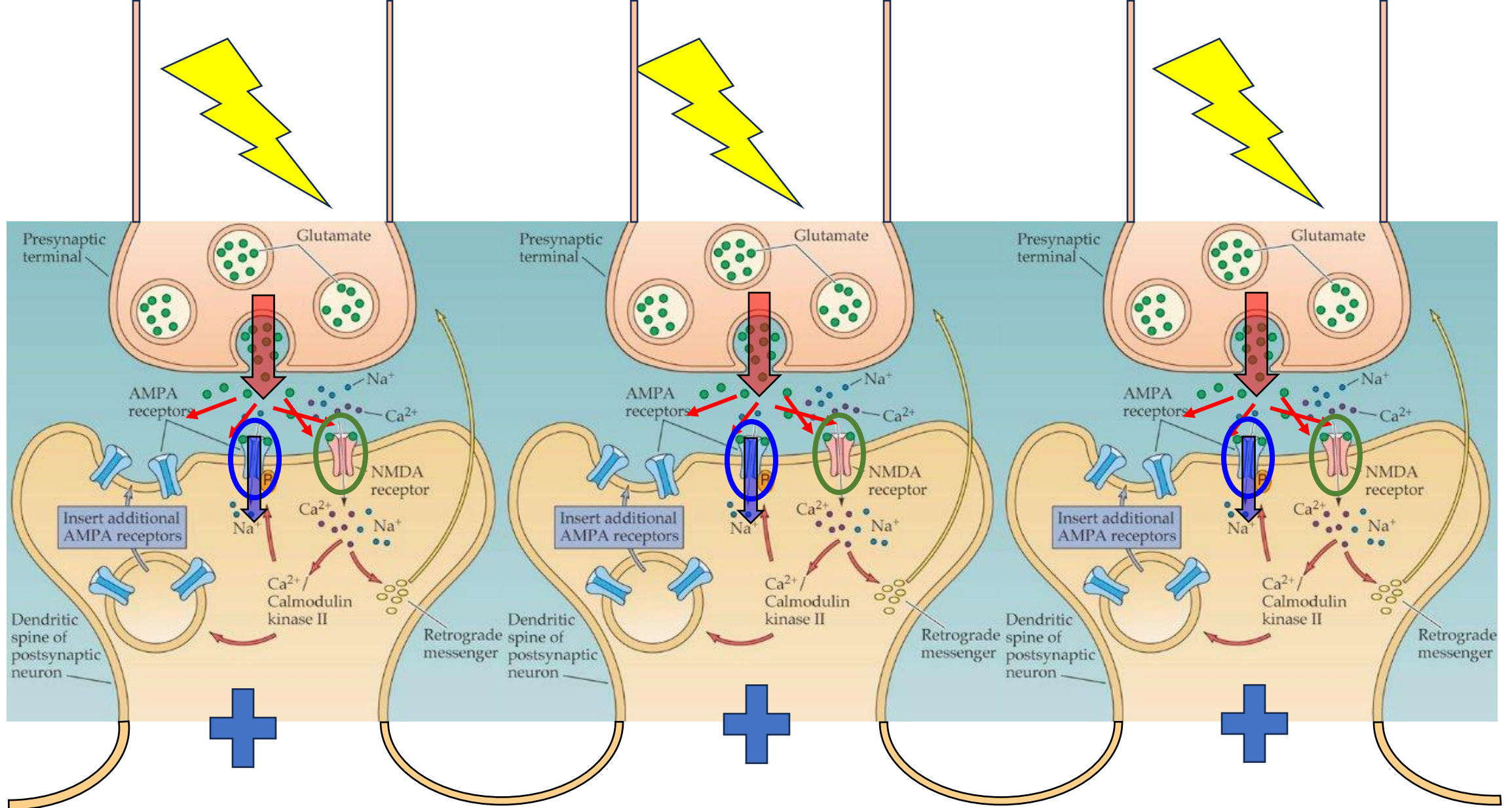
All three presynaptic neurons fire



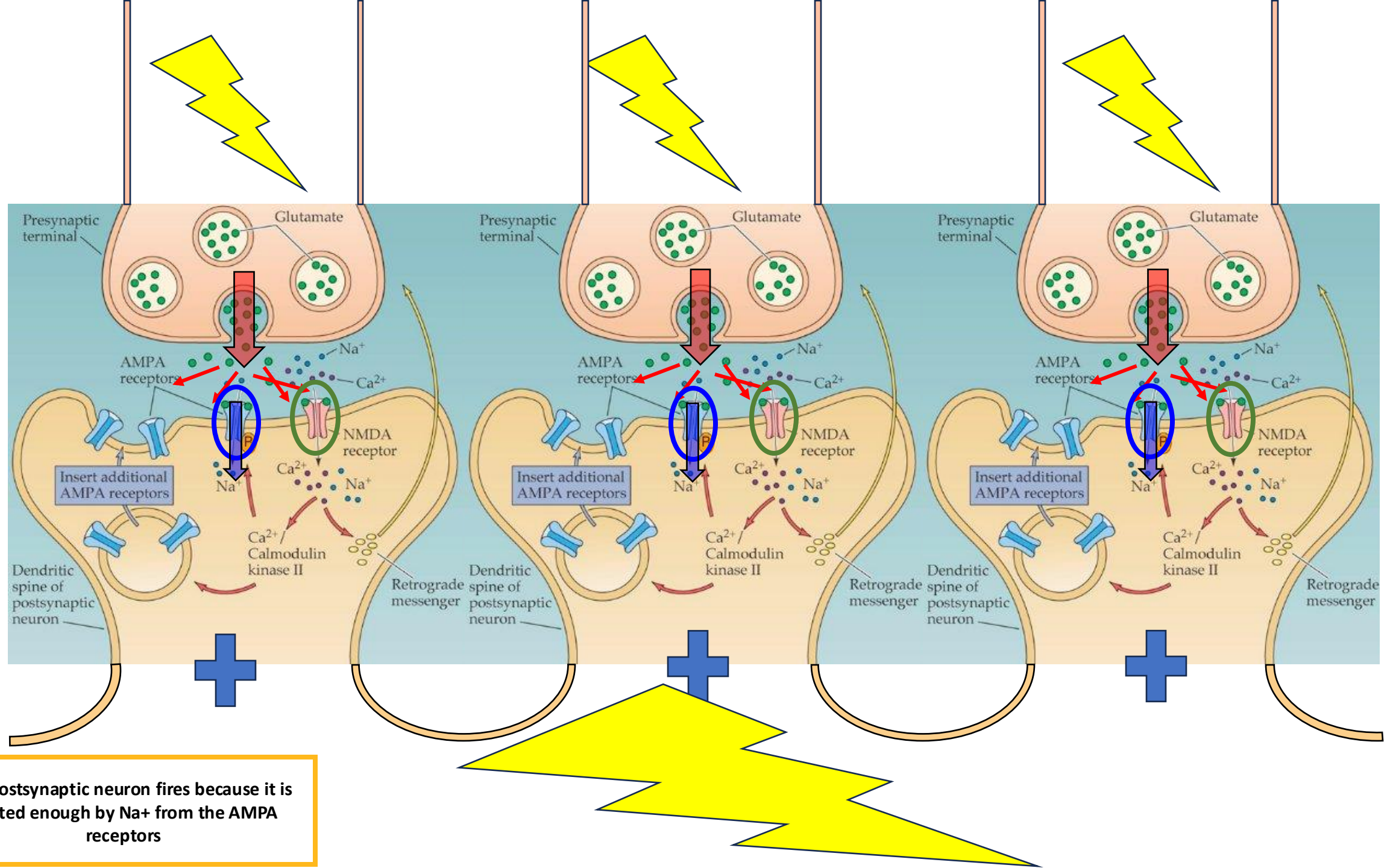
They thus each release Glutamate into the three synapses



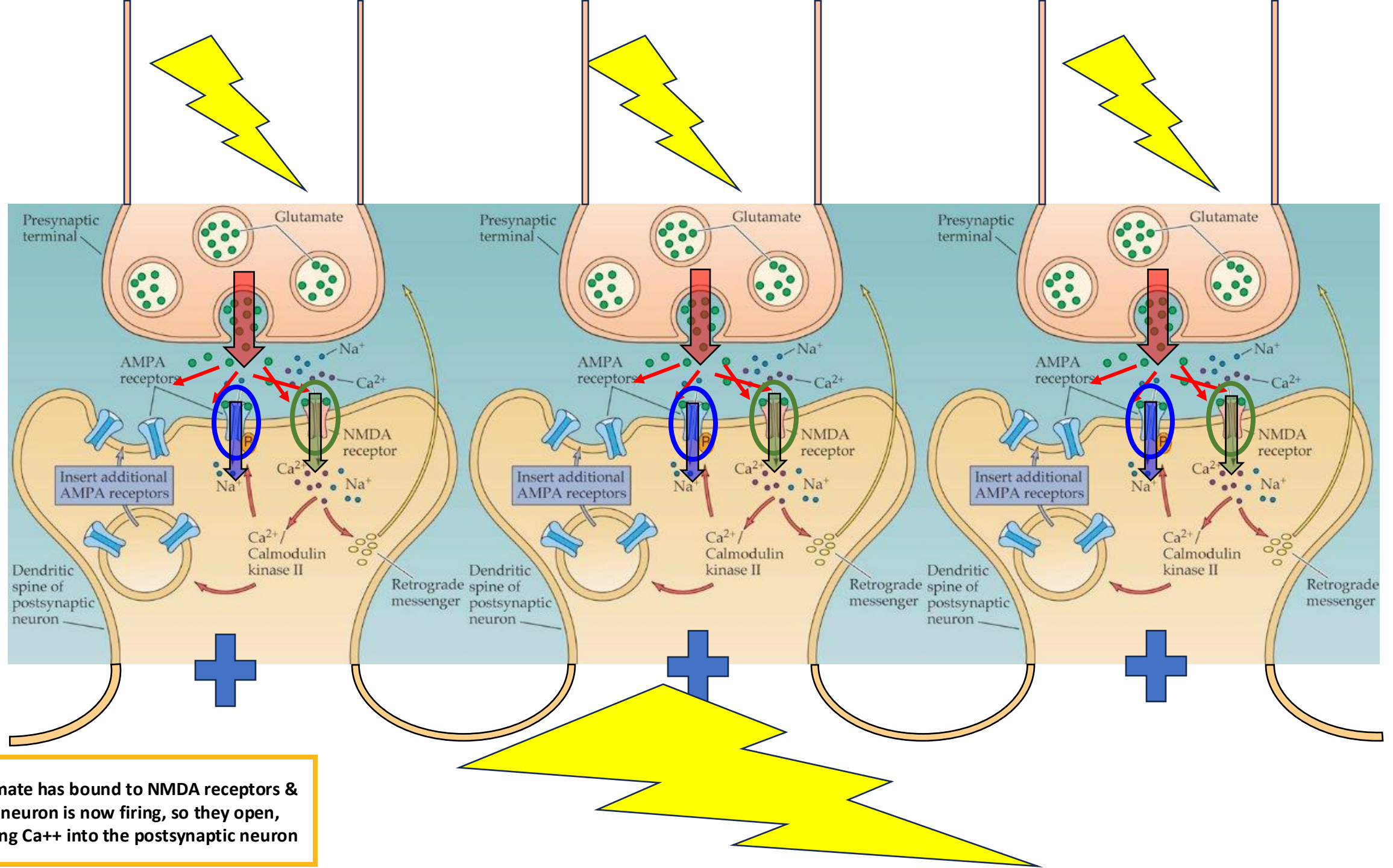
Glutamate binds to AMPA receptors, which open as a result and allow in Na^+ , which excites the postsynaptic neuron



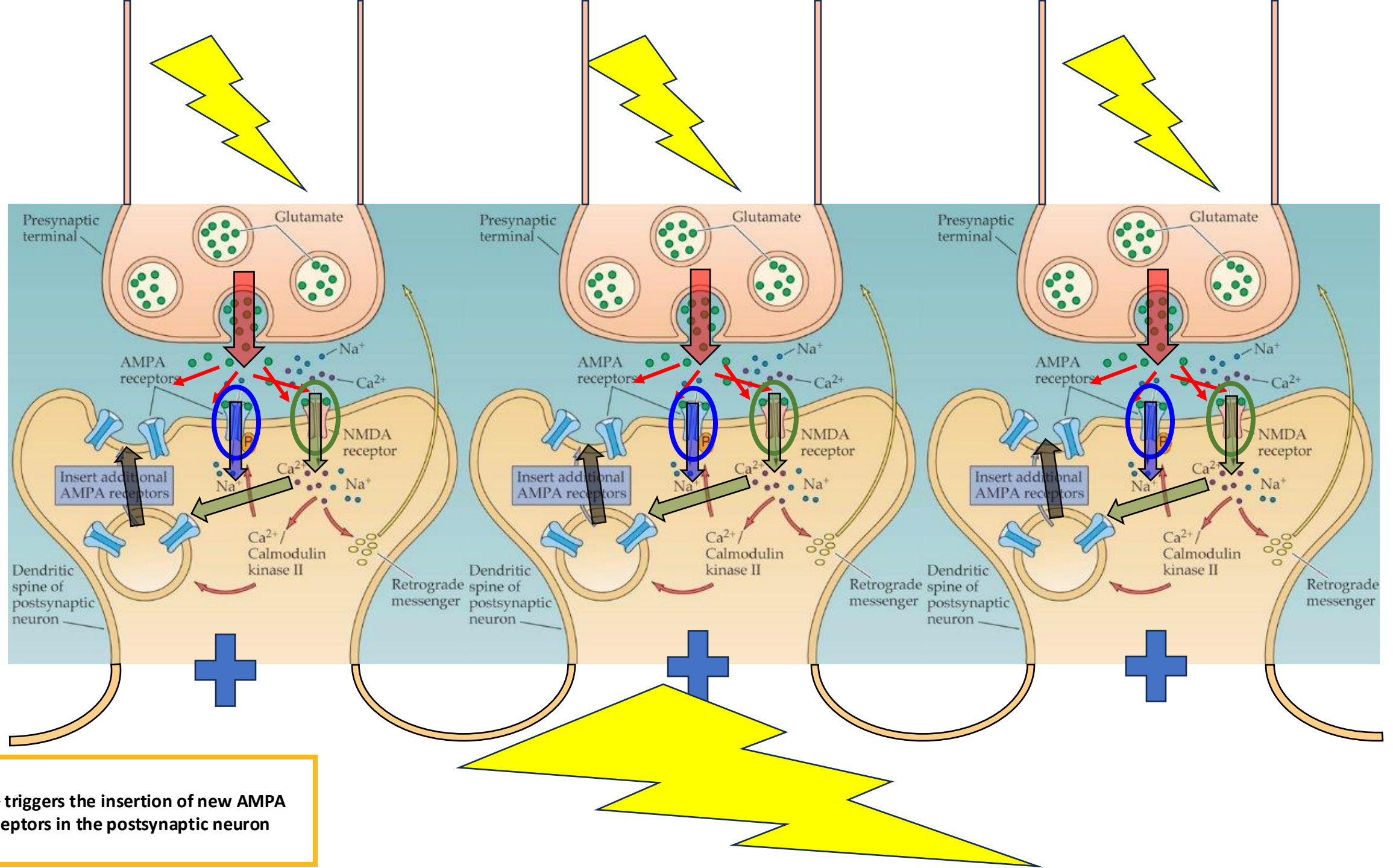
Glutamate has also bound to the NMDA receptors, but nothing happens there



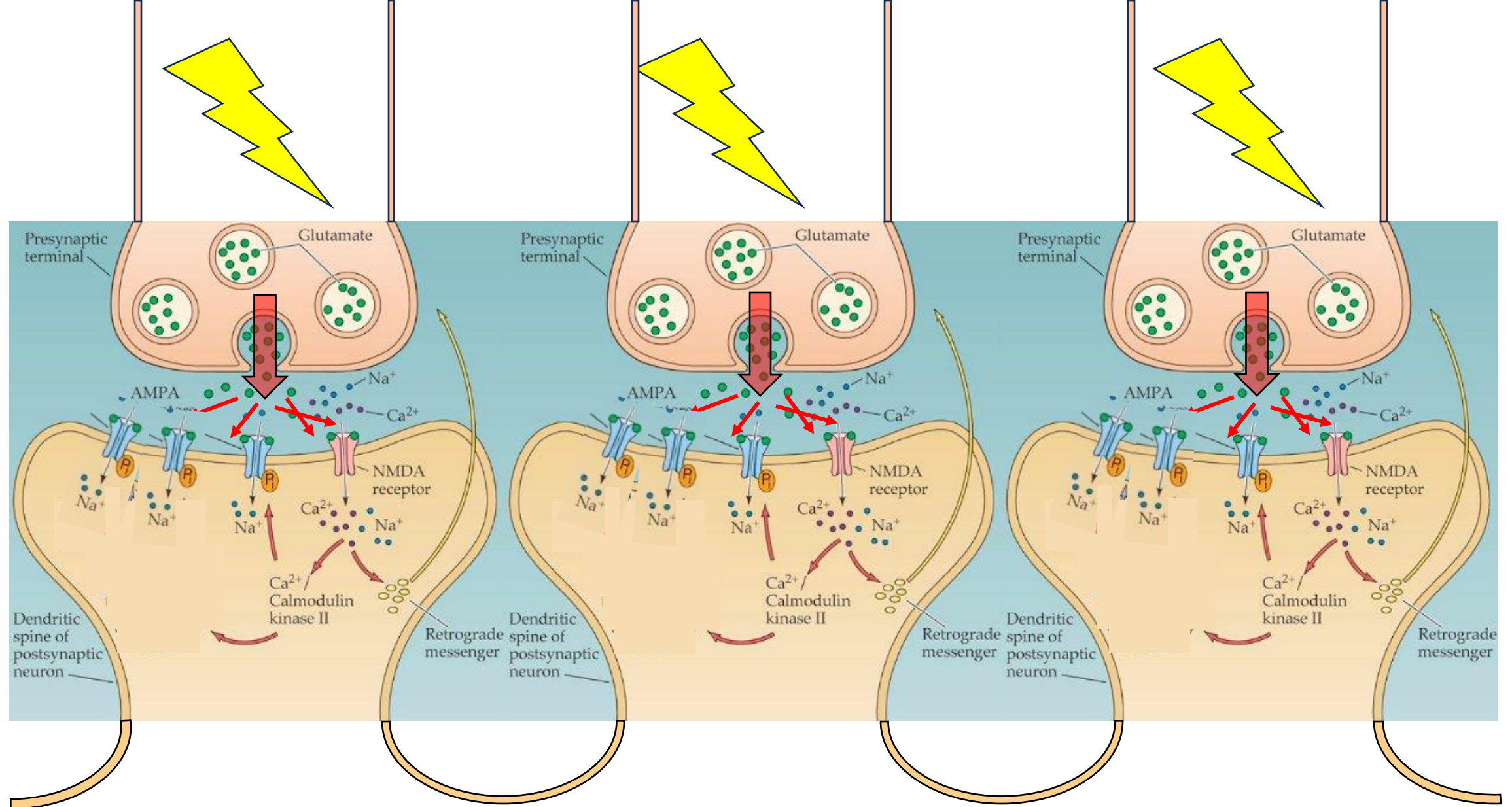
The postsynaptic neuron fires because it is excited enough by Na^+ from the AMPA receptors



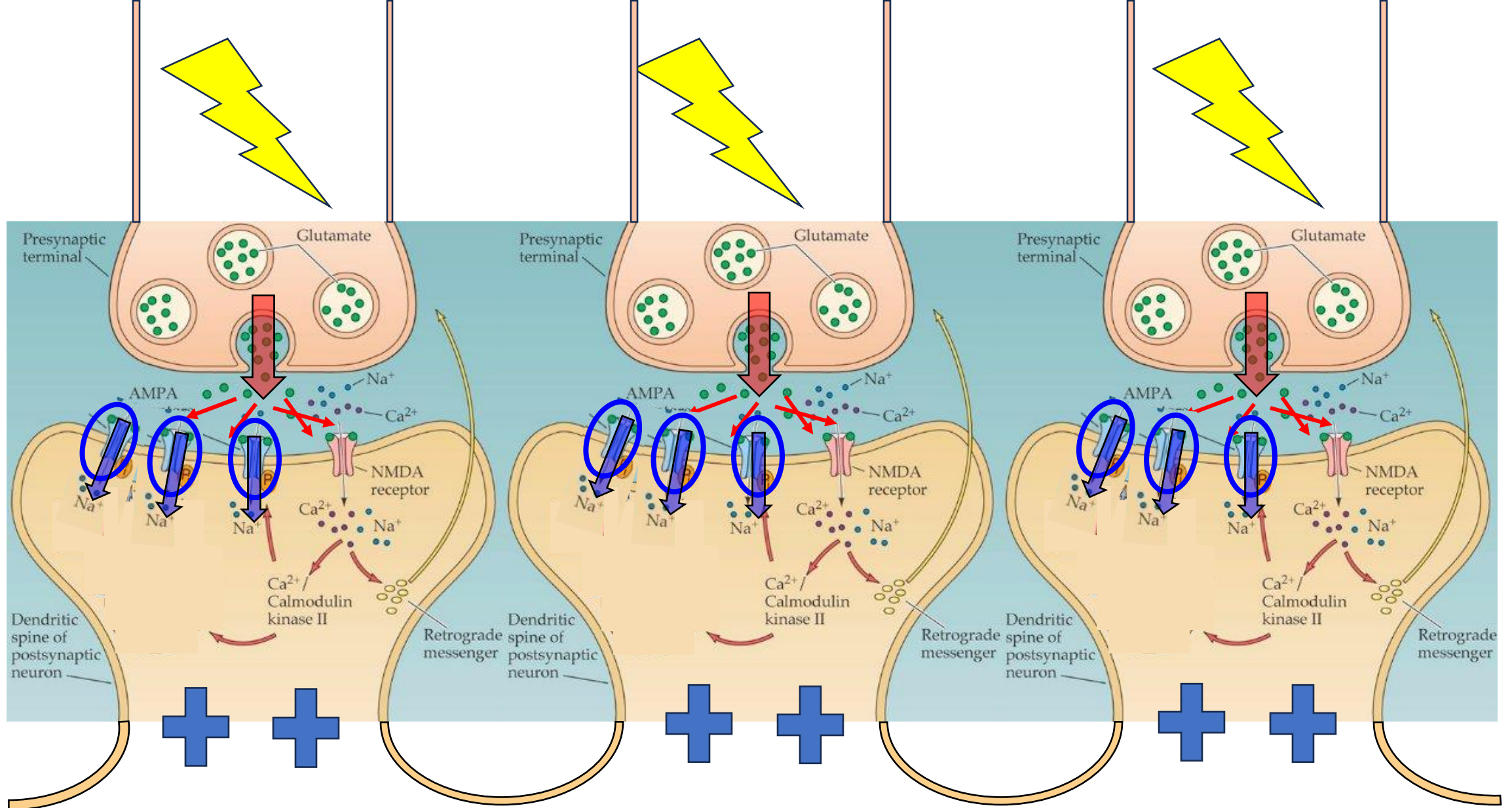
Glutamate has bound to NMDA receptors & this neuron is now firing, so they open, allowing Ca^{++} into the postsynaptic neuron



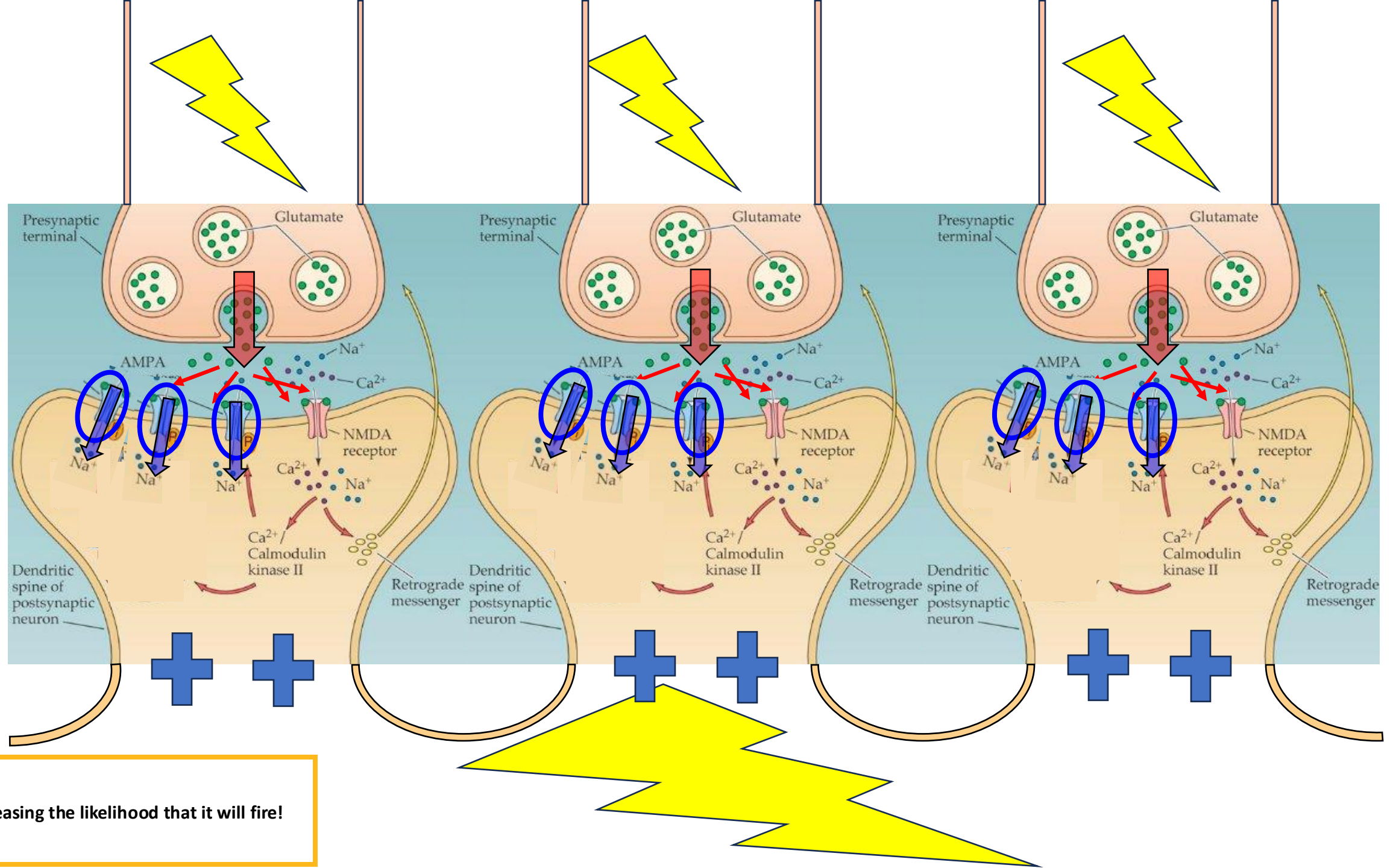
Ca^{2+} triggers the insertion of new AMPA receptors in the postsynaptic neuron

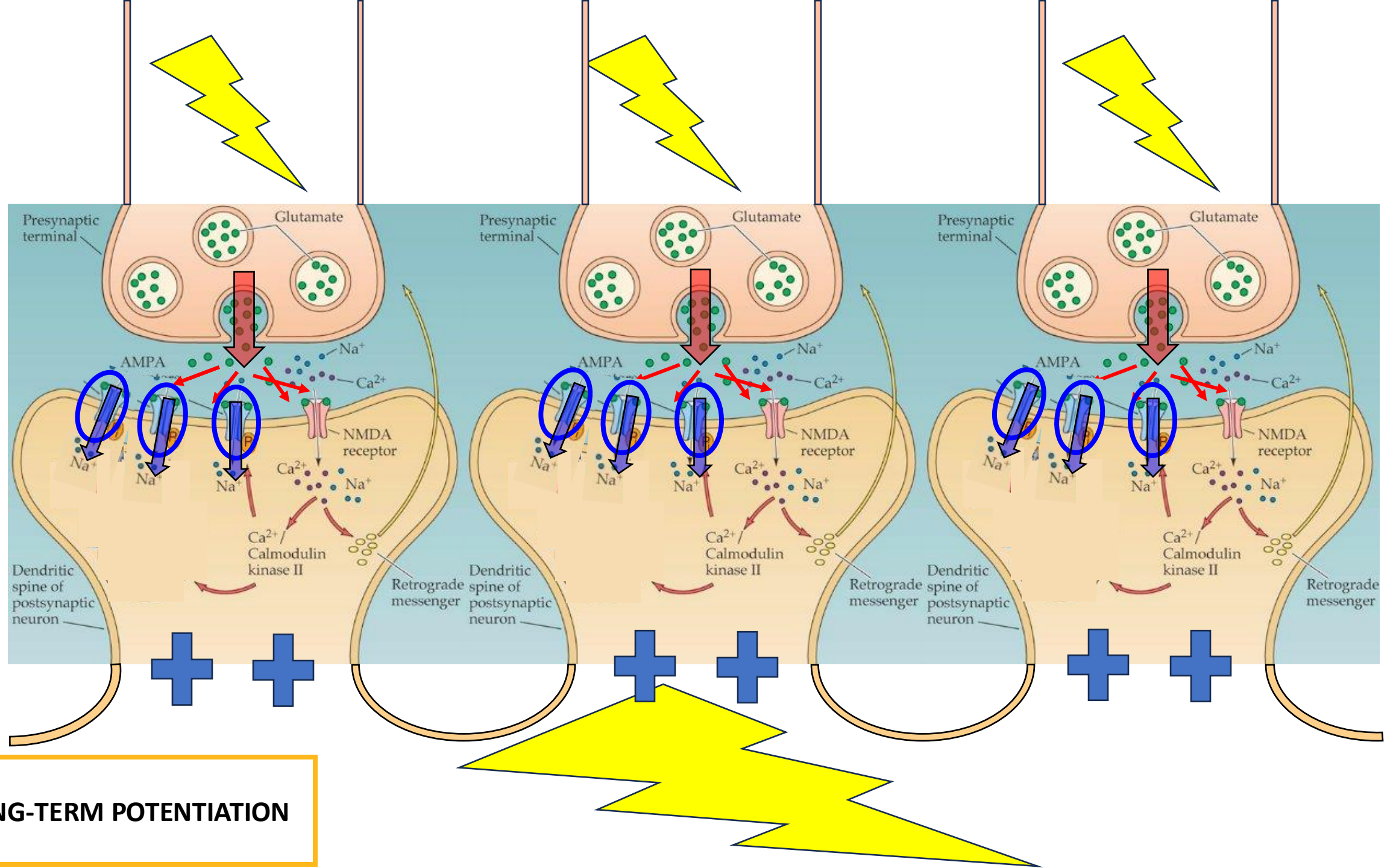


Next time the presynaptic neurons fire & release their Glutamate...

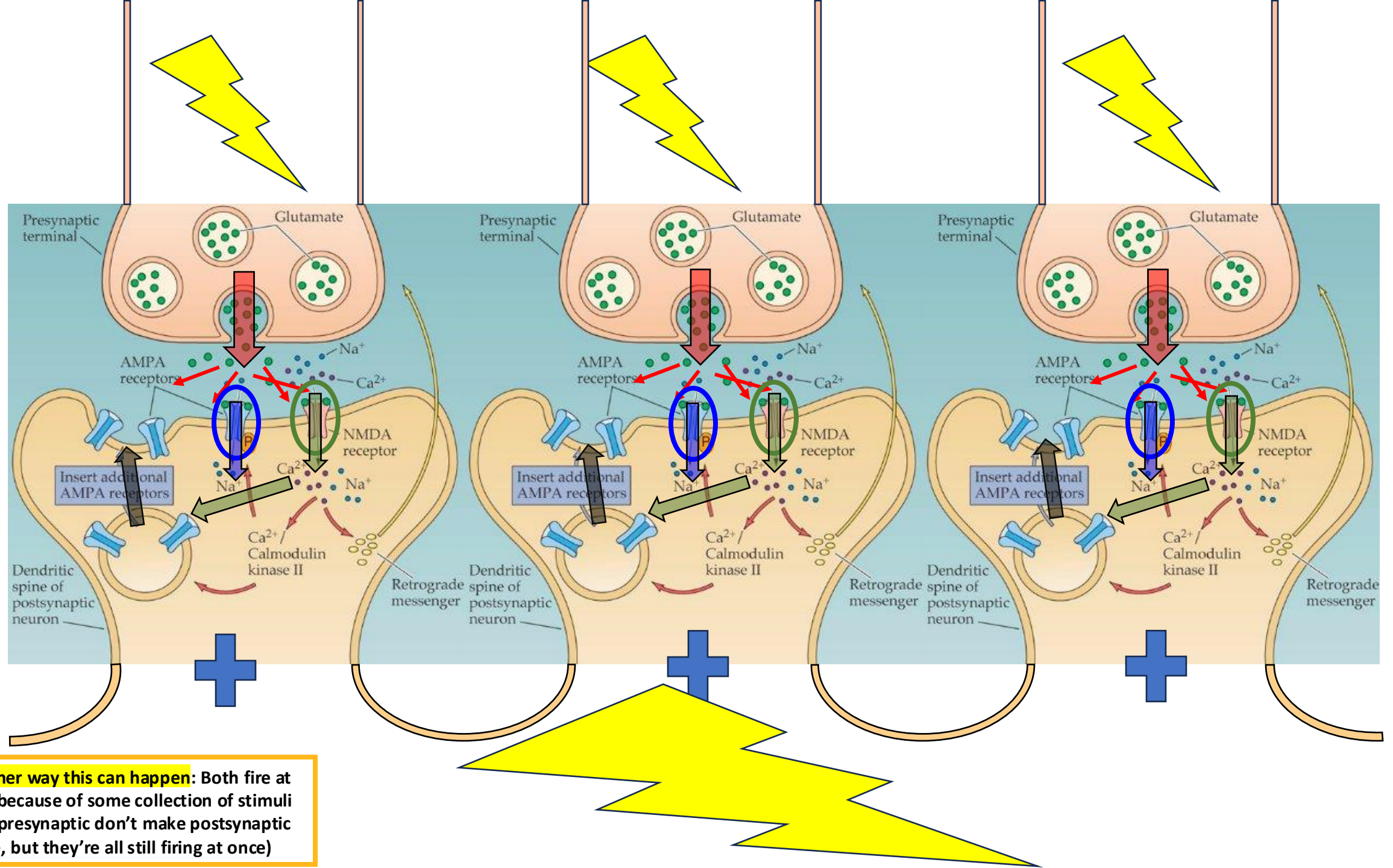


There's more AMPA for the Glutamate to bind to, so more Na^+ enters the postsynaptic neuron and it is more excited...

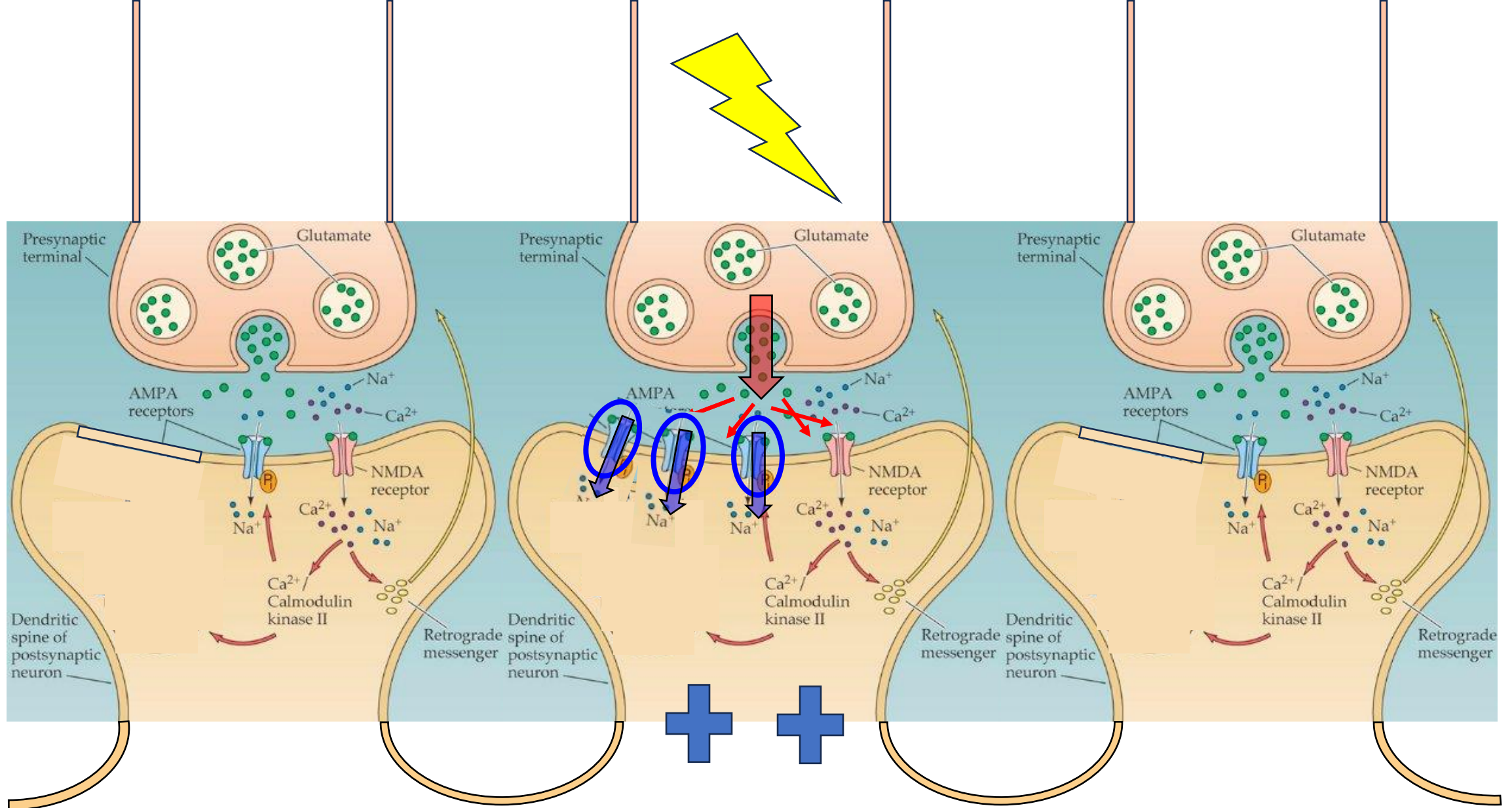




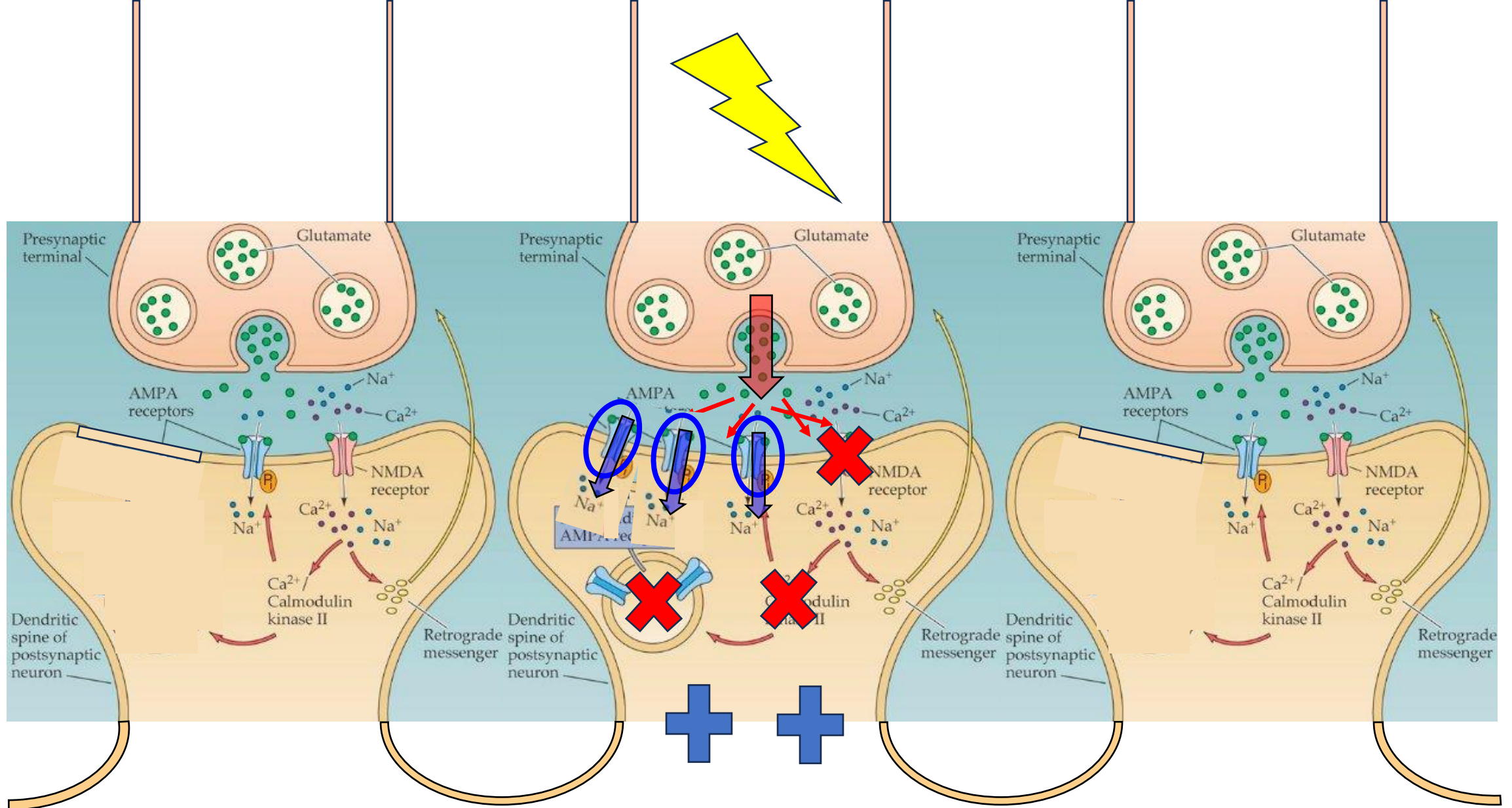
LONG-TERM POTENTIATION



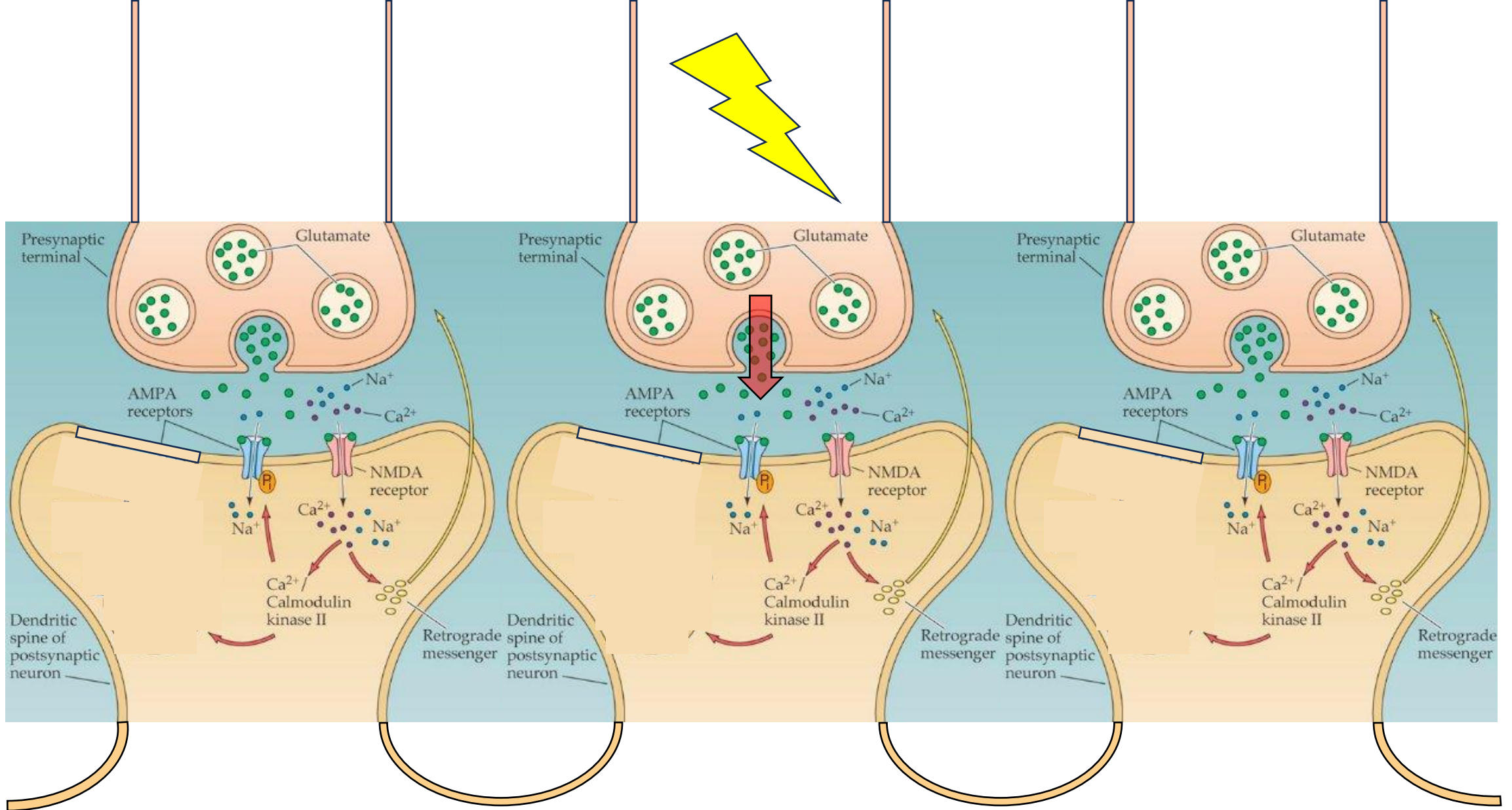
Another way this can happen: Both fire at once because of some collection of stimuli (i.e., presynaptic don't make postsynaptic fire, but they're all still firing at once)



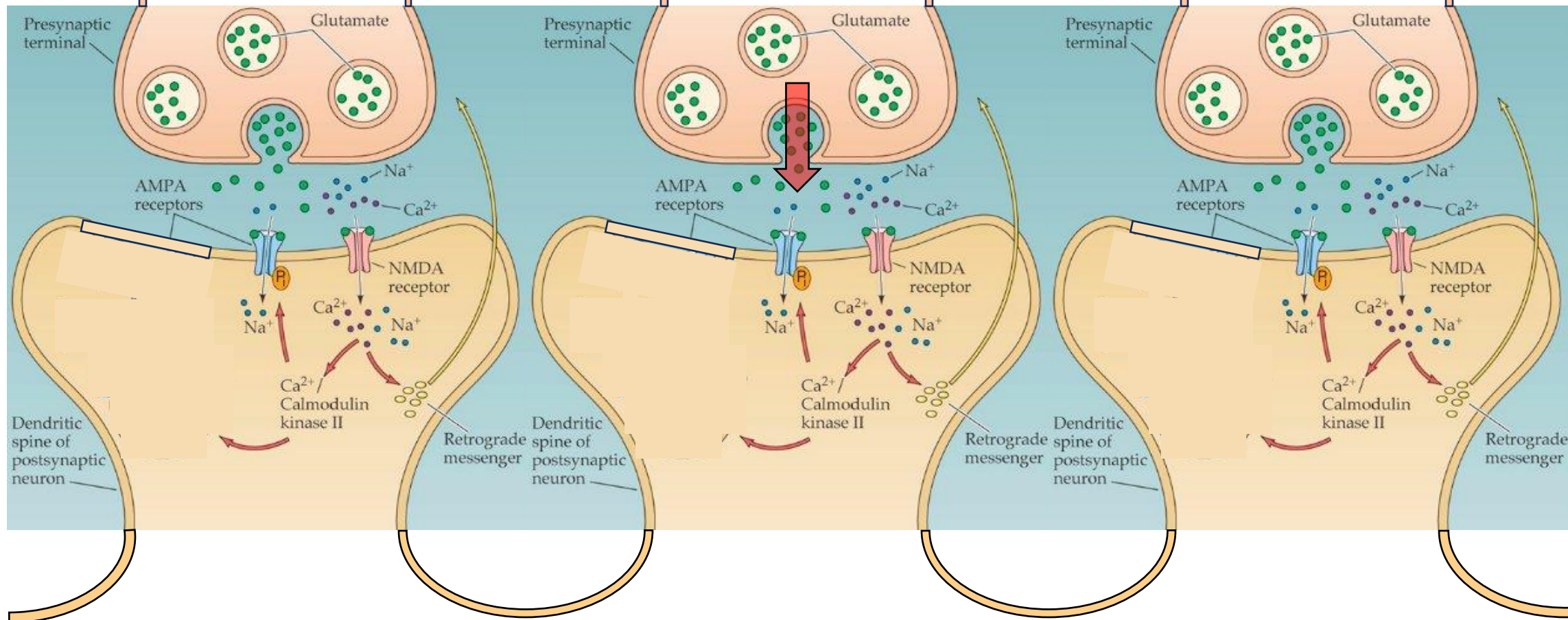
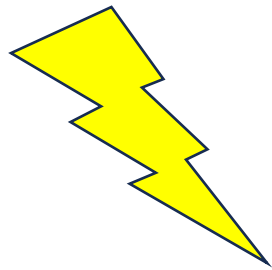
Let's say that a presynaptic neuron fires and its effect isn't powerful enough to trigger the postsynaptic to fire, or that they just aren't activated at the same time...



No fire together, no wire together



If this happens enough, that synapse will *lose* its AMPA – effect of presynaptic on postsynaptic is now weaker

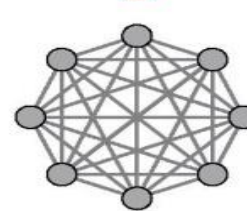
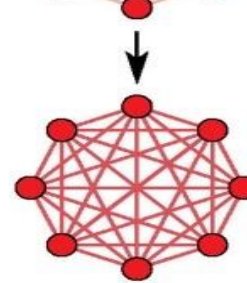
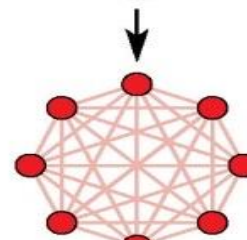
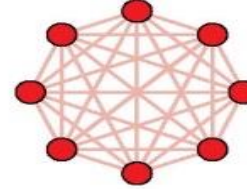


LONG-TERM DEPRESSION

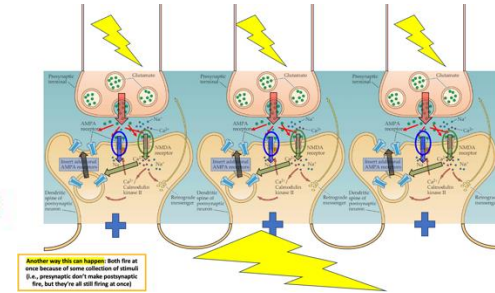
First presentation of stimulus → Neural representation



External stimulus



LONG-TERM POTENTIATION AT GLUTAMATERGIC SYNAPSES OF THE NEURONS INVOLVED IN PROCESSING THIS STIMULUS THROUGHOUT THE BRAIN

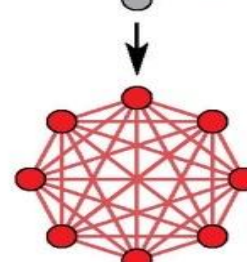
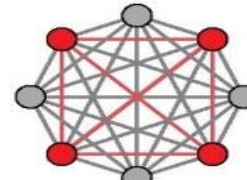


Fire together → Wire together

Wires (strengthened connections) remain after firing stops

TIME PASSES

Later, whole or partial stimulus



Reactivates whole cell assembly

RECAP

Questions?